

Data Mining

Complete Numerical Problems & Solutions

CT 72502 (Elective I)

BEI / BEX · Year IV Part I

Institute of Engineering, Tribhuvan University

Every numerical the course sets, in one volume: **24 distinct methods** and **47 PYQ instances** drawn from **all 17 past papers (2070–2081 BS)**, plus a few extra practice problems where a likely pattern has no paper behind it yet. Each method is explained once, then every paper that used it is solved under the **question exactly as printed**, so a problem can be attempted from this PDF alone. **Every answer was independently recomputed, not copied from the source notes.**

Theory lives in the companion volume, **Data Mining — Complete Exam Notes**.

Ch	Topic	Methods	PYQ solved
2	Data Preprocessing	7	8
3	Classification	6	12
4	Association Analysis	5	14
5	Cluster Analysis	3	11
6	Anomaly / Fraud Detection	2	1
7	Advanced Applications	1	1
Total		24	47

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Ch 2 – Data Preprocessing – Numerical Problems & Solutions

7 distinct numerical methods · 8 PYQ instances solved · 2 extra practice problems. Every answer independently recomputed, not copied from the source notes.

Methods in this chapter

- **M1** Normalisation: min-max, z-score, decimal scaling 78 Bh
- **M2** PCA: principal components + proportion of variance 76 Ch
- **M3** Distance measures + distance matrix (Euclidean, Manhattan, Supremum) 80 Bh, 76 Ch
- **M4** Cosine similarity 80 Bh, 80 Ba, 76 Ch
- **M5** Binary similarity: Jaccard, SMC, symmetric dissimilarity 80 Ba, 76 Ch
- **M6** Schema design + OLAP operation sequence 78 Bh, 76 Ch
- **M7** Central tendency, quartiles, IQR and outliers NEW, no PYQ

M1. Normalisation

Method

- **Min-max** maps v into $[new_min, new_max]$, preserves relationships exactly:

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

- **Z-score** uses mean $\bar{A}$ and std deviation σ_A . Use when min/max unknown, or outliers would squash a min-max result:

$$v' = \frac{v - \bar{A}}{\sigma_A} \quad \sigma_A = \sqrt{\frac{1}{n} \sum (v_i - \bar{A})^2}$$

- **Decimal scaling** moves the decimal point:

$$v' = \frac{v}{10^j}, \quad j = \text{smallest int such that } \max(|v'|) < 1$$

Two traps. (1) Z-score uses the **mean**, not the min. (2) For decimal scaling, j must make the max *strictly* less than 1, so $\max |v| = 1000$ needs $j = 4$, not $j = 3$.

Problem 1.1 2+2+2

QUESTION AS PRINTED 2078 Bhadra · Q3 · [2+2+2]

Use the following methods to normalize the data: 200, 300, 400, 600 and 1000.

- a) Min-max normalization by setting min = 0 and max = 1
- b) Z-score normalization
- c) Normalization by decimal scaling

Given

$v = \{200, 300, 400, 600, 1000\}$, $n = 5$, $min_A = 200$, $max_A = 1000$.

(a) Min-max, $new_min = 0$, $new_max = 1$

$$v' = \frac{v - 200}{1000 - 200} = \frac{v - 200}{800}$$

v	v'
200	$0/800 = 0$
300	$100/800 = 0.125$
400	$200/800 = 0.25$
600	$400/800 = 0.5$
1000	$800/800 = 1$

(c) Decimal scaling

- $\max |v| = 1000$.
- $j = 3$: $1000/10^3 = 1$, not < 1 . **Rejected**.
- $j = 4$: $1000/10^4 = 0.1 < 1$. **Use $j = 4$.**

v	$v' = v/10000$
200	0.02
300	0.03
400	0.04
600	0.06
1000	0.1

Final answer

- Min-max: 0, 0.125, 0.25, 0.5, 1
- Z-score: -1.0607 , -0.7071 , -0.3536 , $+0.3536$, $+1.7678$
- Decimal scaling: 0.02, 0.03, 0.04, 0.06, 0.1

(b) Z-score

- Mean $\bar{A} = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$
- Deviations: -300 , -200 , -100 , $+100$, $+500$
- $\sum (v - \bar{A})^2 = 90000 + 40000 + 10000 + 10000 + 250000 = 400000$
- $\sigma_A = \sqrt{400000/5} = \sqrt{80000} = 282.84$

v	$v' = (v - 500)/282.84$
200	$-300/282.84 = -1.0607$
300	$-200/282.84 = -0.7071$
400	$-100/282.84 = -0.3536$
600	$+100/282.84 = +0.3536$
1000	$+500/282.84 = +1.7678$

Note on σ verified/added

- Above uses the **population** std deviation, $\sqrt{\sum (v - \bar{A})^2 / n}$. This is the form Han & Kamber use, so it is the safe default.
- If the examiner wants the **sample** std deviation, divide by $n - 1$: $s = \sqrt{400000/4} = 316.23$, giving -0.9487 , -0.6325 , -0.3162 , $+0.3162$, $+1.5811$.
- **State which one you used.** Either is accepted if labelled.

Problem 1.2 – all four transformations on one data set 6

QUESTION AS PRINTED 2079 Baishakh · Q2 · [6]

Suppose that the data for analysis include the attribute the frequency of stop words in documents. The values are given in increasing order: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

- Use smoothing by bin means with a depth of 3.
- Use min-max normalization to transform the value 35 into the range from 0.0 to 1.0.
- Use z-score normalization to transform the value 35 where the standard deviation of the above frequency is 12.94.
- Use normalization by decimal scaling to transform the value 35.

$n = 27$, already sorted. “**Depth 3**” means **3 values per bin**, not 3 bins — so $27/3 = 9$ bins. This is the single most common error on this question. Parts (b)–(d) each transform **only the value 35**, so no full table is needed there.

(a) Smoothing by bin means, depth 3

partition in order, replace every value by its bin mean

Bin	Values	Mean	Smoothed to
1	13, 15, 16	$44/3 = 14.67$	14.67, 14.67, 14.67
2	16, 19, 20	$55/3 = 18.33$	18.33, 18.33, 18.33
3	20, 21, 22	$63/3 = 21.00$	21, 21, 21
4	22, 25, 25	$72/3 = 24.00$	24, 24, 24
5	25, 25, 30	$80/3 = 26.67$	26.67, 26.67, 26.67
6	33, 33, 35	$101/3 = 33.67$	33.67, 33.67, 33.67
7	35, 35, 35	$105/3 = 35.00$	35, 35, 35
8	36, 40, 45	$121/3 = 40.33$	40.33, 40.33, 40.33
9	46, 52, 70	$168/3 = 56.00$	56, 56, 56

- **Bin 9 shows why smoothing works**: the outlier 70 is pulled down to 56, and 46 pushed up. Noise is absorbed into the

local average.

- This is **equal-depth** (equal-frequency) binning: same *count* per bin, not the same value range. Compare Practice A, which uses depth 4.

(b), (c), (d): transforming the single value 35

Method	Formula	Substitution	Result
Min-max into $[0, 1]$	$v' = \frac{v - \min}{\max - \min} (new_{max} - new_{min}) + new_{min}$	$\frac{35 - 13}{70 - 13} = \frac{22}{57}$	0.386
Z-score	$v' = \frac{v - \bar{A}}{\sigma_A}$, $\bar{A} = \frac{809}{27} = 29.963$	$\frac{35 - 29.963}{12.94} = \frac{5.037}{12.94}$	0.389
Decimal scaling	$v' = v/10^j$, smallest j with $\max(v') < 1$	$\max = 70 \rightarrow j = 2$, $35/100$	0.35

- σ is given as **12.94**, so do not recompute it – but you *must* compute the mean yourself. Sum the 27 values: 809.
- **Decimal scaling: $j = 2$, not 1.** $j = 1$ would give $70/10 = 7 \not< 1$. Check the *maximum*, not the value being transformed.
- All three land near 0.35–0.39 here, which is coincidence: 35 sits just above the mean and just under the middle of the range.

M2. PCA: principal components and proportion of variance

Method

1. Start from the **covariance matrix** Σ (if given raw data: subtract the mean first, then compute Σ).
2. Solve the characteristic equation $|\Sigma - \lambda I| = 0$ for the eigenvalues λ .
3. Sort λ in **descending** order. λ_i = variance carried by component i .
4. For each λ_i solve $(\Sigma - \lambda_i I) \mathbf{v} = 0$ for the eigenvector, then normalise it to unit length.
5. Proportion of total variance explained by component i :

$$\frac{\lambda_i}{\sum_j \lambda_j}, \quad \sum_j \lambda_j = \text{trace}(\Sigma)$$

- **Check:** the eigenvalues must sum to the trace. Free and catches arithmetic slips.

Problem 2.1 4

QUESTION AS PRINTED 2076 Chaitra · Q1 · [4]

Find the principal components and the proportion of the total variance explained by each when the covariance matrix of the three random variables X_1 , X_2 and X_3 is:

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Step 1: characteristic equation

- The third row/column is decoupled, so Σ splits into a 2×2 block and a 1×1 block.
- From the 1×1 block: $(2 - \lambda) = 0 \Rightarrow \boxed{\lambda = 2}$
- From the 2×2 block:

$$(1 - \lambda)(5 - \lambda) - (-2)(-2) = 0$$

$$5 - 6\lambda + \lambda^2 - 4 = 0 \Rightarrow \lambda^2 - 6\lambda + 1 = 0$$

$$\lambda = \frac{6 \pm \sqrt{36 - 4}}{2} = \frac{6 \pm \sqrt{32}}{2} = 3 \pm 2\sqrt{2}$$

Step 2: eigenvalues, sorted

- $\lambda_1 = 3 + 2\sqrt{2} = \mathbf{5.8284}$
- $\lambda_2 = \mathbf{2}$
- $\lambda_3 = 3 - 2\sqrt{2} = \mathbf{0.1716}$
- **Check:** $5.8284 + 2 + 0.1716 = 8 = \text{trace}(\Sigma) = 1 + 5 + 2$. **Correct.**

Step 3: eigenvectors

- For $\lambda_1 = 5.8284$: row 1 gives $(1 - 5.8284)v_1 - 2v_2 = 0 \Rightarrow v_2 = -2.4142 v_1$; row 3 gives $v_3 = 0$.
 $\mathbf{e}_1 \propto (1, -2.4142, 0)$, length $\sqrt{1 + 5.8284} = 2.6131$
 $\mathbf{e}_1 = (0.3827, -0.9239, 0)$
- For $\lambda_2 = 2$: rows 1 and 2 force $v_1 = v_2 = 0$, v_3 free.
 $\mathbf{e}_2 = (0, 0, 1)$
- For $\lambda_3 = 0.1716$: row 1 gives $0.8284 v_1 = 2v_2 \Rightarrow v_2 = 0.4142 v_1$; $v_3 = 0$.
 $\mathbf{e}_3 \propto (1, 0.4142, 0)$, length 1.0824
 $\mathbf{e}_3 = (0.9239, 0.3827, 0)$
- Check:** $\mathbf{e}_1 \cdot \mathbf{e}_3 = (0.3827)(0.9239) + (-0.9239)(0.3827) = 0$
 $\rightarrow$ orthogonal, as required for a symmetric matrix.

Final answer

- PC1: $\lambda_1 = 5.8284$, $\mathbf{e}_1 = (0.3827, -0.9239, 0)$, **72.86%**
- PC2: $\lambda_2 = 2$, $\mathbf{e}_2 = (0, 0, 1)$, **25.00%**
- PC3: $\lambda_3 = 0.1716$, $\mathbf{e}_3 = (0.9239, 0.3827, 0)$, **2.14%**

Step 4: proportion of variance

PC	λ_i	$\lambda_i/8$
PC1	5.8284	72.86%
PC2	2.0000	25.00%
PC3	0.1716	2.14%
Total	8.0000	100%

Interpretation

- PC1 alone carries $\approx 73\%$ of the variance.
- PC1 + PC2 carry **97.86%** $\rightarrow$ dropping PC3 loses only 2.14%.
- So 3 dims reduce to 2 with almost no information loss.

M3. Distance measures and the distance matrix**Method**

- All three are Minkowski with a different r :

$$\text{dist}(p, q) = \left(\sum_{k=1}^n |p_k - q_k|^r \right)^{1/r}$$

- $r = 1$ **Manhattan**: $\sum |p_k - q_k|$
- $r = 2$ **Euclidean**: $\sqrt{\sum (p_k - q_k)^2}$
- $r \rightarrow \infty$ **Supremum** (L_∞ , Chebyshev): $\max_k |p_k - q_k|$
- A **distance matrix** is symmetric with a zero diagonal, so only the lower triangle needs computing.

Warm-up (Han & Kamber Example 2.19–2.20)

PROBLEM not a PYQ · Han & Kamber Example 2.19–2.20 verified/added

Given the two objects $x_1 = (1, 2)$ and $x_2 = (3, 5)$, compute the Euclidean, Manhattan and Supremum distances between them.

Smallest case that shows all three at once. Worth memorising as a check.

- $x_1 = (1, 2)$, $x_2 = (3, 5)$. Differences: 2 and 3.
- Euclidean = $\sqrt{2^2 + 3^2} = \sqrt{13} = \mathbf{3.61}$
- Manhattan = $2 + 3 = \mathbf{5}$
- Supremum = $\max(2, 3) = \mathbf{3}$
- Ordering**: supremum $\leq$ Euclidean $\leq$ Manhattan. Always true. Use it to catch slips.

Problem 3.1 2

QUESTION AS PRINTED 2076 Chaitra · Q2 (a) · [2]

Given the following points compute the distance matrix using the Manhattan and the Supremum distance.

Points	X	Y
P1	6	3
P2	2	2
P3	3	4

Manhattan (L_1)

- $d(P_1, P_2) = |6 - 2| + |3 - 2| = 4 + 1 = 5$
- $d(P_1, P_3) = |6 - 3| + |3 - 4| = 3 + 1 = 4$
- $d(P_2, P_3) = |2 - 3| + |2 - 4| = 1 + 2 = 3$

Supremum (L_∞)

- $d(P_1, P_2) = \max(4, 1) = 4$
- $d(P_1, P_3) = \max(3, 1) = 3$
- $d(P_2, P_3) = \max(1, 2) = 2$

	P_1	P_2	P_3
P_1	0	5	4
P_2	5	0	3
P_3	4	3	0

	P_1	P_2	P_3
P_1	0	4	3
P_2	4	0	2
P_3	3	2	0

- **Sanity check:** supremum $\leq$ Euclidean $\leq$ Manhattan, always. Holds here ($4 \leq 5$, $3 \leq 4$, $2 \leq 3$).

Problem 3.2 3

QUESTION AS PRINTED 2080 Bhadra · Q1 (b) · [3+3+3]

How do similarity / dissimilarity is calculated? Find the cosine similarity between Object-2 and 4. Also calculate the Euclidean distance between object 1, 3 and object 1, 4.

Object	Size	Weight	Color_Code	Taste_Score
1	4	56	7	10
2	3	53	8	11
3	7	58	6	9
4	9	55	7	12

The cosine-similarity part of this same question is Problem 4.1 below.

Step-by-step

- $d(O_1, O_3)$: differences $(4 - 7, 56 - 58, 7 - 6, 10 - 9) = (-3, -2, 1, 1)$
 $= \sqrt{9 + 4 + 1 + 1} = \sqrt{15} = \mathbf{3.873}$
- $d(O_1, O_4)$: differences $(4 - 9, 56 - 55, 7 - 7, 10 - 12) = (-5, 1, 0, -2)$
 $= \sqrt{25 + 1 + 0 + 4} = \sqrt{30} = \mathbf{5.477}$

Final answer

$d(O_1, O_3) = 3.873$, $d(O_1, O_4) = 5.477$. Object 1 is closer to object 3 than to object 4.

M4. Cosine similarity**Method**

$$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \|d_2\|}, \quad d_1 \cdot d_2 = \sum_k d_{1k} d_{2k}, \quad \|d\| = \sqrt{\sum_k d_k^2}$$

- Measures **angle only**, ignores magnitude $\rightarrow$ used for documents.
- Always lay out 3 lines: dot product, then each length, then divide.

Typo in your class notes verified/added : DBK's Lecture-2 cosine example prints $\|d_2\| = (6)^{0.5} = 2.245$. In fact $\sqrt{6} = \mathbf{2.449}$. His final answer 0.3150 is correct and only follows from $2.449 (5 / (6.481 \times 2.449) = 0.315)$, so 2.245 is a slip. Copy the method, not that number.

Problem 4.1 3

QUESTION AS PRINTED 2080 Bhadra · Q1 (b) · [3+3+3]

How do similarity / dissimilarity is calculated? Find the cosine similarity between Object-2 and 4. Also calculate the Euclidean distance between object 1, 3 and object 1, 4.

Object	Size	Weight	Color_Code	Taste_Score
1	4	56	7	10
2	3	53	8	11
3	7	58	6	9
4	9	55	7	12

The Euclidean-distance part of this same question is Problem 3.2 above.

Given

$$O_2 = (3, 53, 8, 11), \quad O_4 = (9, 55, 7, 12).$$

Step-by-step

- Dot product: $3(9) + 53(55) + 8(7) + 11(12) = 27 + 2915 + 56 + 132 = 3130$
- $\|O_2\| = \sqrt{9 + 2809 + 64 + 121} = \sqrt{3003} = 54.800$
- $\|O_4\| = \sqrt{81 + 3025 + 49 + 144} = \sqrt{3299} = 57.437$
- $\cos = \frac{3130}{54.800 \times 57.437} = \frac{3130}{3147.5}$

Final answer

$$\cos(O_2, O_4) = \mathbf{0.9944}.$$

- Nearly 1 $\rightarrow$ the two objects point almost the same direction.
- **Why so high:** *Weight* (≈ 50) dwarfs the other attributes and dominates the dot product. **Normalise before using cosine** on mixed-scale attributes, or the answer means very little. verified/added

Problem 4.2 1

QUESTION AS PRINTED 2080 Baishakh · Q2 (b)(iv) · [1]

Find the Cosine similarity between documents $d_1 = (4, 1, 2, 0, 2, 0, 0)$ and $d_2 = (2, 1, 3, 0, 1, 1, 1)$. (the rest of Q2 b is Problem 5.2 below)

- Dot: $8 + 1 + 6 + 0 + 2 + 0 + 0 = 17$
- $\|d_1\| = \sqrt{16 + 1 + 4 + 0 + 4 + 0 + 0} = \sqrt{25} = 5$
- $\|d_2\| = \sqrt{4 + 1 + 9 + 0 + 1 + 1 + 1} = \sqrt{17} = 4.1231$
- $\cos = 17 / (5 \times 4.1231) = 17 / 20.6155$

Final answer

0.8246

Problem 4.3 1

QUESTION AS PRINTED 2076 Chaitra · Q2 (b) · [1]

Given the following two vectors compute the Cosine similarity between them.

$$D_1 = [4 \ 0 \ 2 \ 0 \ 1] \quad D_2 = [2 \ 0 \ 0 \ 2 \ 2]$$

- Dot: $8 + 0 + 0 + 0 + 2 = 10$
- $\|D_1\| = \sqrt{16 + 0 + 4 + 0 + 1} = \sqrt{21} = 4.5826$
- $\|D_2\| = \sqrt{4 + 0 + 0 + 4 + 4} = \sqrt{12} = 3.4641$
- $\cos = 10 / (4.5826 \times 3.4641) = 10 / 15.8745$

Final answer

0.6300

M5. Binary similarity: Jaccard, SMC, symmetric dissimilarity

Method

- Encode each object as a 0/1 vector, then count over all attribute positions:
 - M_{11} : both 1 M_{00} : both 0
 - M_{10} : $p = 1, q = 0$ M_{01} : $p = 0, q = 1$
- SMC (symmetric binary, 0-0 counts): $\frac{M_{11} + M_{00}}{M_{11} + M_{00} + M_{10} + M_{01}}$
- Jaccard (asymmetric binary, 0-0 ignored): $\frac{M_{11}}{M_{11} + M_{10} + M_{01}}$
- Symmetric binary dissimilarity: $d = \frac{M_{10} + M_{01}}{M_{11} + M_{00} + M_{10} + M_{01}} = 1 - SMC$
- Nominal (non-binary) dissimilarity: $d = (p - m) / p$, $m = \text{matches}$, $p = \text{total attrs}$.

Problem 5.1 2

QUESTION AS PRINTED 2076 Chaitra · Q2 (c) · [2]

Given the following two binary vectors compute the Jaccard similarity and Simple Matching Coefficient.

$$P = [0 \ 0 \ 1 \ 1 \ 0 \ 1] \quad Q = [1 \ 1 \ 1 \ 1 \ 0 \ 1]$$

Step-by-step: tabulate position by position

Position	1	2	3	4	5	6
P	0	0	1	1	0	1
Q	1	1	1	1	0	1
Category	M_{01}	M_{01}	M_{11}	M_{11}	M_{00}	M_{11}

- $M_{11} = 3, \quad M_{01} = 2, \quad M_{10} = 0, \quad M_{00} = 1 \quad (\text{total} = 6 \checkmark)$
- $J = \frac{3}{3+0+2} = \frac{3}{5}$
- $SMC = \frac{3+1}{6} = \frac{4}{6}$

Final answer

$J = 0.6, \quad SMC = 0.667$. SMC is higher because it credits the one 0-0 match that Jaccard discards.

Problem 5.2 2+2+2

QUESTION AS PRINTED 2080 Baishakh · Q2 (b) · [2+2+2+1]

Consider the following table:

Name	Gender	Eyecolor	Haircolor	Test-1	Test-2	Fever	Cough
Ram	M	Black	Gray	P	N	P	N
Laxmi	F	Blue	Black	P	P	N	N
Shyam	M	Blue	Gray	N	P	N	P

- Calculated Jaccard Coefficient of $Sim_{jaccard}$ (Ram, Laxmi) for asymmetric binary attributes.
- Dissimilarity of symmetric binary attributes d (Laxmi, Shyam).
- Find the Simple matching coefficients of SMC (Ram, Shyam).
- Find the Cosine similarity between documents $d_1 = (4, 1, 2, 0, 2, 0, 0)$ and $d_2 = (2, 1, 3, 0, 1, 1, 1)$ — solved as Problem 4.2 above.

Step 0: split the attributes (the whole question turns on this)

- **Asymmetric binary** = Test-1, Test-2, Fever, Cough. Encode $P = 1, N = 0$. A shared “N” is uninformative, which is exactly what “asymmetric” means.
- **Symmetric** = Gender, Eyecolor, Haircolor. Both/all states are equally meaningful.
- Encoded: Ram = (1, 0, 1, 0), Laxmi = (1, 1, 0, 0), Shyam = (0, 1, 0, 1).

(i) Jaccard(Ram, Laxmi)

	T1	T2	Fev	Cou
Ram	1	0	1	0
Laxmi	1	1	0	0
Cat.	M_{11}	M_{01}	M_{10}	M_{00}

$$- M_{11} = 1, M_{01} = 1, M_{10} = 1, M_{00} = 1$$

$$- J = \frac{1}{1+1+1} = \frac{1}{3}$$

Answer

$$J = 0.333$$

(iii) SMC(Ram, Shyam)

	T1	T2	Fev	Cou
Ram	1	0	1	0
Shyam	0	1	0	1
Cat.	M_{10}	M_{01}	M_{10}	M_{01}

$$- M_{11} = 0, M_{00} = 0, M_{10} = 2, M_{01} = 2$$

$$- SMC = \frac{0+0}{4} = 0$$

Answer

$SMC = 0$ (they disagree on every test).

(ii) $d(\text{Laxmi, Shyam})$, symmetric binary attributes

- Across these 3 people each of Gender / Eyecolor / Haircolor takes exactly **2 states**, and neither state is an “absence” → all 3 are **symmetric binary**.
- Encode: Gender $M=1$ · Eyecolor Black=1 · Haircolor Gray=1.

	Gender	Eye	Hair
Laxmi (F, Blue, Black)	0	0	0
Shyam (M, Blue, Gray)	1	0	1
Category	s	t	s

$$- q = 0, r = 0, s = 2, t = 1 \quad (\text{total } p = 3 \checkmark)$$

- Symmetric binary dissimilarity keeps t :

$$d = \frac{r+s}{q+r+s+t} = \frac{0+2}{3} = \frac{2}{3}$$

Answer

$$d = 0.667$$

Cross-check: for binary attrs the symmetric formula reduces to the nominal mismatch ratio $(p-m)/p = (3-1)/3 = 2/3$. Same answer by both routes, so the encoding choice is safe.

verified/added

M6. Schema design + OLAP operation sequence

Method

1. Put the **measures** in a central **fact table**, plus one foreign key per dimension.
2. Draw one **dimension table** per dimension around it.
 - **Star** → leave dimension tables flat (de-normalised).
 - **Snowflake** → normalise the hierarchies into sub-tables (location → city → province).
3. For the OLAP sequence, work from the base cuboid and ask, per dimension:
 - Need a coarser level? → **roll-up**
 - Need a finer level? → **drill-down**
 - Restricting 1 dimension? → **slice**. Restricting ≥ 2 ? → **dice**
 - Dimension not wanted at all? → **roll-up to all** (dimension reduction)
4. n dimensions → 2^n cuboids in the lattice.

Problem 6.1 3+3

QUESTION AS PRINTED 2078 Bhadra · Q2 · [3+3]

Suppose that a data warehouse consists of the four dimensions data, spectator, location, and game, and the two measures count and charge, where charge is the fare that a spectator pays when watching a game on a given date. Spectators may be students, adults or seniors, with each category having its own charge rate.

- a) Draw a star schema diagram for the data warehouse.
- b) Starting with the base cuboid [data, spectator, location, game], what specific OLAP operations should you perform in order to list the total charge paid by student spectators at Dashrath Stadium in 2021?

(a) Star schema

- **Fact table** Sales: date_key, spectator_key, location_key, game_key, count, charge.
- **Dimension tables** (flat, one hop from the fact table):
 - date(date_key, day, month, quarter, year)
 - spectator(spectator_key, name, status, phone, address) where status $\in$ {student, adult, senior}
 - location(location_key, name, street, city, province, country)
 - game(game_key, name, description, sport_type)
- 4 dimensions $\rightarrow 2^4 = 16$ cuboids in the lattice.

(b) OLAP operation sequence

1. **Roll-up** on *date* from date_key to year.
 2. **Roll-up** on *spectator* from spectator_key to status.
 3. **Roll-up** on *location* from location_key to location_name.
 4. **Roll-up** on *game* to all (the question does not split by game).
 5. **Dice** with year = 2021 and status = student and location_name = ``Dashrath Stadium''.
 6. Read off the charge measure (summed).
- **Dice, not slice** $\rightarrow$ 3 dimensions are being restricted at once.

Problem 6.2 3+3

QUESTION AS PRINTED 2076 Chaitra · Q3 · [3+3]

Suppose that a data warehouse for a sales company consists of five dimensions: *time*, *location*, *supplier*, *brand*, and *product*, and two measures: *count* and *price*.

- a) Draw a *snowflake schema* diagram for the data warehouse.
- b) Starting with the base cuboid [*time*, *location*, *supplier*, *brand*, *product*], what specific OLAP operations should one perform in order to list the total *count* for a certain *brand* for each *state* per year (assume *location* has three levels: *country*, *state*, *city*; and assume *time* has three levels: *year*, *month*, *day*)?

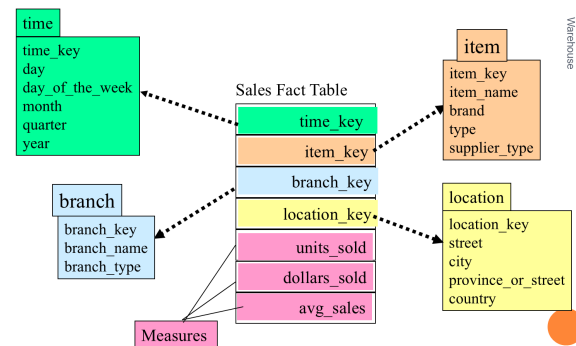
(a) Snowflake schema

- **Fact table** Sales: time_key, location_key, supplier_key, brand_key, product_key, count, price.
- **Normalised hierarchies** (this is what makes it a snowflake, not a star):
 - time(time_key, day, month_key) $\rightarrow$ month(month_key, month, year)
 - location(location_key, city_key) $\rightarrow$ city(city_key, city, state_key) $\rightarrow$ state(state_key, state, country)
 - product(product_key, name, brand_key) $\rightarrow$ brand(branch_key, brand_name)
 - supplier(supplier_key, supplier_name, supplier_type)
- 5 dimensions $\rightarrow 2^5 = 32$ cuboids.

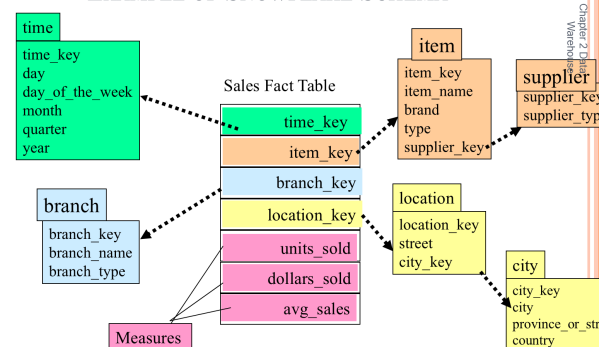
(b) OLAP operation sequence

1. **Roll-up** on *time* from day to year.
2. **Roll-up** on *location* from city to state.
3. **Roll-up** on *supplier* to all (not wanted).
4. **Roll-up** on *product* to brand level.
5. **Slice** on brand = the given brand (1 dimension restricted).
6. Result: cuboid [year, state] holding total count.

EXAMPLE OF STAR SCHEMA



EXAMPLE OF SNOWFLAKE SCHEMA



M7. Central tendency, quartiles, IQR and outliers

NEW Not in any of the 23 papers. Added because Notes by PC/unit 2 Numerical.docx and Untitled spreadsheet.xlsx (2025–26) are built entirely around it. Theory in Ch 2, §2.5.

Method

- **Grouped median** = $L_1 + \left(\frac{N/2 - (\sum freq)_l}{freq_{median}} \right) \times width$. Find the median interval first: the one where cumulative frequency first reaches $N/2$.
- **Mode from mean and median** $\approx 3 median - 2 mean$ (unimodal, moderately skewed).
- **Quartiles**, interpolated convention: position of $Q_k = \frac{k(n+1)}{4}$, then interpolate between the neighbouring sorted values.
- **Outlier fences**: below $Q_1 - 1.5 IQR$ or above $Q_3 + 1.5 IQR$.

Problem 7.1 – grouped median, 100 employees 4

PROBLEM not a PYQ · PC's notes · [4]

Suppose the salaries of 100 employees of a company are grouped into \$10,000 intervals as shown in the table below. Find the median salary.

Salary (\$)	Freq	Cum freq
10,000–20,000	8	8
20,000–30,000	15	23
30,000–40,000	20	43
40,000–50,000	32	75
50,000–60,000	18	93
60,000–70,000	7	100

Solution

1. $N = 100 \rightarrow N/2 = 50$. Need the 50th observation.
2. Cumulative frequency passes 50 inside **40k–50k** (43 before it, 75 after) $\rightarrow$ that is the median interval.
3. $L_1 = 40,000$, $(\sum freq)_l = 43$,
 $freq_{median} = 75 - 43 = 32$, $width = 10,000$.
4. $median = 40,000 + \frac{50-43}{32} \times 10,000 = 40,000 + \frac{7}{32} \times 10,000$

Answer

$median = \$42,187.50$

Problem 7.2 – same method, smaller table 3

PROBLEM not a PYQ · PC's notes · [3]

Salaries (in \$1000s) of 60 employees fall in the intervals 10–20, 20–30, 30–40, 40–50, 50–60 and 60–70 with frequencies 5, 8, 12, 20, 10 and 5 respectively. Find the median salary.

- $N = 5 + 8 + 12 + 20 + 10 + 5 = 60 \rightarrow N/2 = 30$.
- Cumulative before 40–50 = $5 + 8 + 12 = 25$; $freq_{median} = 20$, $L_1 = 40$, $width = 10$.
- $median = 40 + \frac{30-25}{20} \times 10 = 40 + 2.5 = 42.5$ (i.e. \$42,500).

Problem 7.3 – mode from mean and median 2

PROBLEM not a PYQ · PC's notes · [2]

For a unimodal, moderately skewed distribution the mean is 60 and the median is 55. Estimate the mode, and state what the result says about the skew.

- $mean - mode \approx 3(mean - median) \Rightarrow mode \approx 3median - 2mean$.
- $mode \approx 3(55) - 2(60) = 165 - 120 = 45$.
- **Sanity check**: $mode < median < mean \rightarrow$ positively skewed. Consistent.

Problem 7.4 – full descriptive summary of one dataset 2+2+2+2

PROBLEM not a PYQ · Han & Kamber Example 2.6 · [2+2+2+2]

Suppose that the data for analysis is the salaries (in \$1000s) of 12 employees, already sorted in increasing order:

30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110

Find (a) the mean, median and mode; (b) the quartiles Q_1 and Q_3 and the IQR; (c) the five-number summary; (d) any outliers using the $1.5 \times IQR$ rule.

(a) Central tendency

- $\sum x = 696 \rightarrow \text{mean} = \frac{696}{12} = 58.$
- $n \text{ even} \rightarrow \text{median} = \frac{52+56}{2} = 54.$
- 52 and 70 each occur twice $\rightarrow$ **bimodal**, modes **52** and **70**.
- $\text{midrange} = \frac{30+110}{2} = 70.$
- $\text{mode} < \text{median} < \text{mean}$ is broken here by the bimodality; the single outlier 110 is what pulls the mean above the median.

(b) Quartiles, positional convention (Han & Kamber)

- Take the 3rd, 6th, 9th sorted values $\rightarrow Q_1 = 47, Q_2 = 52, Q_3 = 63.$
- $IQR = 63 - 47 = 16.$

(c) Quartiles, interpolated convention (PC's class notes)

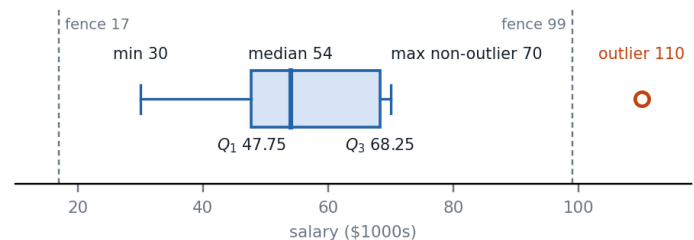
- $Q_1 \text{ at } \frac{1(13)}{4} = 3.25 \rightarrow 47 + 0.25(50 - 47) = 47.75$
- $Q_2 \text{ at } \frac{2(13)}{4} = 6.5 \rightarrow 52 + 0.5(56 - 52) = 54$
- $Q_3 \text{ at } \frac{3(13)}{4} = 9.75 \rightarrow 63 + 0.75(70 - 63) = 68.25$
- $IQR = 68.25 - 47.75 = 20.5$

(d) Five-number summary and outliers

- Five-number summary (interpolated): **30, 47.75, 54, 68.25, 110.**
- Fences: $47.75 - 1.5(20.5) = 17.0$ and $68.25 + 1.5(20.5) = 99.0.$
- $110 > 99 \rightarrow$ **110** is a suspected outlier. Nothing below 17.
- **Cross-check with the other convention:** fences $47 - 24 = 23$ and $63 + 24 = 87.$ **110 is still the only outlier** $\rightarrow$ the verdict does not depend on the convention, so either is safe in the exam as long as you name it.

(e) Boxplot

- Box from Q_1 to Q_3 , line at 54, whiskers to 30 and to the largest non-outlier (70), and **110 plotted as a lone point.**



Drawn to scale by `boxfig.py` from the values above. The whisker stops at 70, not at the fence.

Extra practice (patterns not yet fully covered)**Practice A – binning for noise smoothing**

Binning is examined in prose 72 Ka, 76 Ash but never as a numerical. **This is DBK's own worked example** (Lecture-1, "Binning Methods for Data Smoothing"), so it is the likeliest form any future numerical would take. Answers below independently recomputed and they match his slide exactly.

PROBLEM not a PYQ · DBK Lecture-1 worked example verified/added

The following price data has already been sorted:

4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

Partition the data into **equal-depth** bins of depth 4, then smooth the noise by (i) bin means and (ii) bin boundaries.

Solution

- 12 values, depth 4 $\rightarrow$ 3 bins.
 - Bin 1: 4, 8, 9, 15 Bin 2: 21, 21, 24, 25 Bin 3: 26, 28, 29, 34
- **(i) Smoothing by bin means**
 - Bin 1 mean = $36/4 = 9 \rightarrow 9, 9, 9, 9$
 - Bin 2 mean = $91/4 = 22.75 \approx 23 \rightarrow 23, 23, 23, 23$
 - Bin 3 mean = $117/4 = 29.25 \approx 29 \rightarrow 29, 29, 29, 29$
- **(ii) Smoothing by bin boundaries** (each value $\rightarrow$ nearer of min/max)
 - Bin 1 (min 4, max 15): 4, 8, 9, 15 $\rightarrow 4, 4, 4, 15$ (8 and 9 are nearer 4)
 - Bin 2 (min 21, max 25): 21, 21, 24, 25 $\rightarrow 21, 21, 25, 25$
 - Bin 3 (min 26, max 34): 26, 28, 29, 34 $\rightarrow 26, 26, 26, 34$
- **Equal-width** for contrast: range = $34 - 4 = 30, N = 3 \rightarrow W = (B - A)/N = 10.$ Intervals $[4, 14), [14, 24), [24, 34] \rightarrow$

counts 3, 3, 6. **Unequal counts: that is the difference from equal-depth.**

Practice B — min-max into a range other than [0,1]

Every PYQ so far uses $[0, 1]$. The figures below are Han & Kamber 3rd ed Example 3.4, so they are the canonical ones. verified/added

PROBLEM not a PYQ · Han & Kamber 3rd ed Example 3.4 verified/added

Suppose that the minimum and maximum values for the attribute *income* are \$12,000 and \$98,000 respectively. Normalise **\$73,600** by min-max normalization, first to the range $[0, 1]$ and then to the range $[-1, 1]$.

Solution

- To $[0, 1]$: $v' = \frac{73600 - 12000}{98000 - 12000}(1 - 0) + 0 = \frac{61600}{86000} = \mathbf{0.716}$
- To $[-1, 1]$: $v' = \frac{61600}{86000}(1 - (-1)) + (-1) = 0.716(2) - 1 = \mathbf{0.432}$
- **Check:** the value sits 71.6% of the way up the range, so in $[-1, 1]$ it must land slightly above the midpoint 0. It does.

Ch 3 – Classification – Numerical Problems & Solutions

6 distinct methods · 12 PYQ instances solved · every answer independently recomputed. Decision-tree and Naive-Bayes figures cross-checked against Han & Kamber 3rd ed Examples 8.1–8.4, which use the same dataset the IOE papers copy.

Methods

- **M1** Decision tree by information gain (ID3) 79 Bh, 78 Bh, 76 Ch
- **M2** Gini index 75 Ch, 71 Ch
- **M3** Naive Bayes classification 72 Ch, 79 Bh
- **M4** k -Nearest Neighbour 80 Bh
- **M5** Confusion matrix metrics **8 papers**
- **M6** ANN gradients (backpropagation) 76 Ch

M1. Decision tree by information gain (ID3)

Method

1. Compute $Info(D) = -\sum p_i \log_2 p_i$ for the whole set.
 2. For each attribute A : split D , compute $Info_A(D) = \sum \frac{|D_j|}{|D|} Info(D_j)$, then $Gain(A) = Info(D) - Info_A(D)$.
 3. **Highest gain wins** → becomes the node's test.
 4. Recurse on each branch. Stop when a branch is **pure** ($Info = 0$).
- **Time-savers**: a pure partition contributes 0; a 50/50 partition contributes exactly 1; two partitions with the same class ratio have the same $Info$, so compute once.
 - $\log_2 x = \ln x / \ln 2$. Keep 4 decimals until the end.

Problem 1.1 10

QUESTION AS PRINTED 2079 Bhadra · Q3 · [10]

Draw decision tree for the given data using ID3 algorithm.

Age	Income	Student	Credit_Rating	Buy's_Computer
Youth	High	No	Fair	No
Youth	High	No	Excellent	No
Middle_Aged	High	No	Fair	Yes
Senior	Medium	No	Fair	Yes
Senior	Low	Yes	Fair	Yes
Senior	Low	Yes	Excellent	No
Middle_Aged	Low	Yes	Excellent	Yes
Youth	Medium	No	Fair	No
Youth	Low	Yes	Fair	Yes
Senior	Medium	Yes	Fair	Yes
Youth	Medium	Yes	Excellent	Yes
Middle_Aged	Medium	No	Excellent	Yes
Middle_Aged	High	Yes	Fair	Yes
Senior	Medium	No	Excellent	No

Given

14 tuples. Class counts: **Yes = 9, No = 5**.

Step 1: entropy of the whole set

$$Info(D) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.4098 + 0.5305 = \mathbf{0.940}$$

Step 2: gain of each attribute

Attribute	Partition (Yes, No)	$Info_A(D)$	Gain
Age	Youth (2,3) · Middle_Aged (4,0) · Senior (3,2)	$\frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971) = 0.694$	0.246
Income	High (2,2) · Medium (4,2) · Low (3,1)	$\frac{4}{14}(1.0) + \frac{6}{14}(0.918) + \frac{4}{14}(0.811) = 0.911$	0.029
Student	Yes (6,1) · No (3,4)	$\frac{7}{14}(0.592) + \frac{7}{14}(0.985) = 0.788$	0.151
Credit_Rating	Fair (6,2) · Excellent (3,3)	$\frac{8}{14}(0.811) + \frac{6}{14}(1.0) = 0.892$	0.048

– **Age has the highest gain → Age is the root.**

– Note Middle_Aged is (4, 0): **pure**, so that branch is immediately a leaf “Yes”. That purity is what makes Age win.

Step 3: recurse on the Youth branch

(5 tuples: 2 Yes, 3 No, $Info = 0.971$)

- Student = Yes → (2 Yes, 0 No) Student = No → (0 Yes, 3 No)
- Both pure → $Info_{Student} = 0$ → $Gain = 0.971$, the maximum possible.
- **Student** is the test under Youth.

Step 4: recurse on the Senior branch

(5 tuples: 3 Yes, 2 No, $Info = 0.971$)

- Credit = Fair → (3 Yes, 0 No) Credit = Excellent → (0 Yes, 2 No)
- Both pure → $Gain = 0.971$. **Credit_Rating** is the test under Senior.

Final tree

- **Age?**
 - Youth → **Student?**
 - no → **No**
 - yes → **Yes**
 - Middle_Aged → **Yes**
 - Senior → **Credit_Rating?**
 - fair → **Yes**
 - excellent → **No**
- Tree has **5 leaves**, depth 2, and classifies all 14 training tuples correctly.

8.2 Decision Tree Induction 331

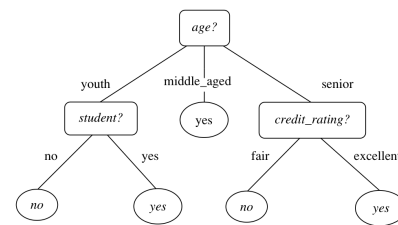


Figure 8.2 A decision tree for the concept *buys.computer*, indicating whether an *AlIElectronics* customer is likely to purchase a computer. Each internal (nonleaf) node represents a test on

The same tree as printed in Han & Kamber 3rd ed, Fig 8.2. Our independently computed gains match their Example 8.1 exactly.

Problem 1.2 8

QUESTION AS PRINTED 2078 Bhadra · Q4 · [8]

Construct a decision tree for the following data set using information gain.

Predict the class label for a data point with values <Female, 2, standard, high>

Gender	Car ownership	Travel cost	Income level	Transport mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car
Female	1	Cheap	Medium	Train
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train

Given

Classes: Bus = 4, Car = 3, Train = 3. $n = 10$.

$$Info(D) = -\frac{4}{10} \log_2 \frac{4}{10} - \frac{3}{10} \log_2 \frac{3}{10} - \frac{3}{10} \log_2 \frac{3}{10} = 0.5288 + 0.5211 + 0.5211 = \mathbf{1.571}$$

Step 2: gain of each attribute

partitions written as (Bus, Car, Train)

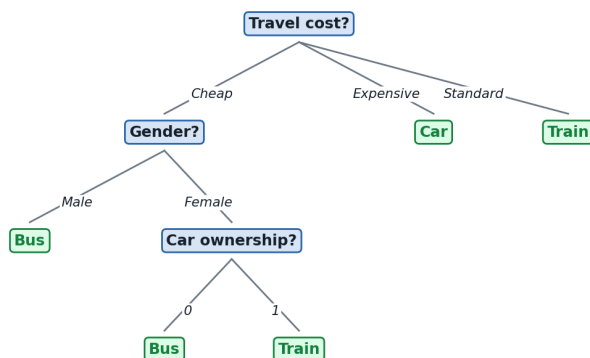
Attribute	Partition (Bus, Car, Train)	$Info_A(D)$	Gain
Travel cost	Cheap (4,0,1) · Expensive (0,3,0) · Standard (0,0,2)	$\frac{5}{10}(0.722) + \frac{3}{10}(0) + \frac{2}{10}(0) = 0.361$	1.210
Income level	Low (2,0,0) · Medium (2,1,3) · High (0,2,0)	$\frac{2}{10}(0) + \frac{6}{10}(1.459) + \frac{2}{10}(0) = 0.875$	0.695
Car ownership	0 (2,0,1) · 1 (2,1,2) · 2 (0,2,0)	$\frac{3}{10}(0.918) + \frac{5}{10}(1.522) + \frac{2}{10}(0) = 1.036$	0.534
Gender	Male (3,1,1) · Female (1,2,2)	$\frac{5}{10}(1.371) + \frac{5}{10}(1.522) = 1.446$	0.125

- **Travel cost has the highest gain → it is the root**, and two of its three branches (Expensive, Standard) are already pure.
- **Only the Cheap branch needs splitting**: 5 tuples, (4 Bus, 0 Car, 1 Train), $Info = 0.722$.

Step 3: recurse on the Cheap branch5 tuples, $Info = 0.722$

Attribute	Partition (Bus, Train)	$Info_A$	Gain
Gender	Male (3,0) pure · Female (1,1)	$\frac{3}{5}(0) + \frac{2}{5}(1.0) = 0.400$	0.322
Car ownership	0 (2,0) pure · 1 (2,1)	$\frac{2}{5}(0) + \frac{3}{5}(0.918) = 0.551$	0.171
Income level	Low (2,0) pure · Medium (2,1)	$\frac{2}{5}(0) + \frac{3}{5}(0.918) = 0.551$	0.171

- **Gender wins**. Male is pure Bus; *Female* still holds (1 Bus, 1 Train).
- That last pair splits perfectly on **Car ownership** (0 → Bus, 1 → Train), gain = 1.0. **Income level splits it equally well** — say so and pick one, either is correct.

Final tree**Prediction for <Female, 2, standard, high>**

- Read the root first: Travel cost = **standard**.
- That branch is **pure Train**, so the walk stops immediately.
- **No other attribute is consulted** — Gender, car ownership and income are never reached. **Answer: Train**.
- **This is the point of the question**: a decision tree only tests the attributes on the path it takes, not all of them.

5 leaves, depth 3, classifies all 10 training tuples correctly. Drawn by `dtfig.py`, which reruns ID3 on the table and checks the gains and the tree against this page.

Problem 1.3 2+9+3

QUESTION AS PRINTED 2082 Bhadra · Q3 · [2+9+3]

When do we use decision tree? Construct a complete decision tree using ID-3 algorithm for following dataset. How do you evaluate the performance of this model?

SN	Credit history	Debt	Collateral	Income	Credit Risk
1	bad	high	none	\$0 to \$15K	high
2	unknown	high	none	\$15 to \$35K	high
3	unknown	low	none	\$15 to \$35K	moderate
4	unknown	low	none	\$0 to \$15K	high
5	unknown	low	none	over \$35K	low
6	unknown	low	adequate	over \$35K	low
7	bad	low	none	\$0 to \$15K	high
8	bad	low	adequate	over \$35K	moderate
9	good	low	none	over \$35K	low
10	good	high	adequate	over \$35K	low
11	good	high	none	\$0 to \$15K	high
12	good	high	none	\$15 to \$35K	moderate
13	good	high	none	over \$35K	low
14	bad	high	none	\$15 to \$35K	high

Three classes, not two, so $Info$ has three terms. Classes: high = 6, moderate = 3, low = 5, $n = 14$.

Step 1: entropy of the whole set

$$Info(D) = -\frac{6}{14} \log_2 \frac{6}{14} - \frac{3}{14} \log_2 \frac{3}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.5239 + 0.4741 + 0.5305 = \mathbf{1.531}$$

Step 2: gain of each attribute

partitions written as (high, moderate, low)

Attribute	Partition (high, mod, low)	$Info_A(D)$	Gain
Income	\$0–15K (4,0,0) pure · \$15–35K (2,2,0) · over \$35K (0,1,5)	$\frac{4}{14}(0) + \frac{4}{14}(1.0) + \frac{6}{14}(0.650) = 0.564$	0.966
Credit history	bad (3,1,0) · unknown (2,1,2) · good (1,1,3)	$\frac{4}{14}(0.811) + \frac{5}{14}(1.522) + \frac{5}{14}(1.371) = 1.265$	0.266
Collateral	none (6,2,3) · adequate (0,1,2)	$\frac{11}{14}(1.435) + \frac{3}{14}(0.918) = 1.325$	0.206
Debt	high (4,1,2) · low (2,2,3)	$\frac{7}{14}(1.379) + \frac{7}{14}(1.557) = 1.468$	0.063

– **Income wins by a wide margin** → **it is the root**, and its \$0–15K branch is already pure (all high risk).

Step 3: recurse on \$15–35K

4 tuples (2 high, 2 moderate), $Info = 1.0$

Attribute	$Info_A$	Gain
Credit history	$\frac{2}{4}(1.0) + \frac{1}{4}(0) + \frac{1}{4}(0) = 0.500$	0.500
Debt	$\frac{3}{4}(0.918) + \frac{1}{4}(0) = 0.689$	0.311
Collateral	all “none” → 1.000	0.000

- **Credit history** splits it: good → moderate, bad → high, **unknown is still mixed** (rows 2 and 3).
- That last pair splits perfectly on **Debt**: high → high risk, low → moderate. Gain = 1.0.

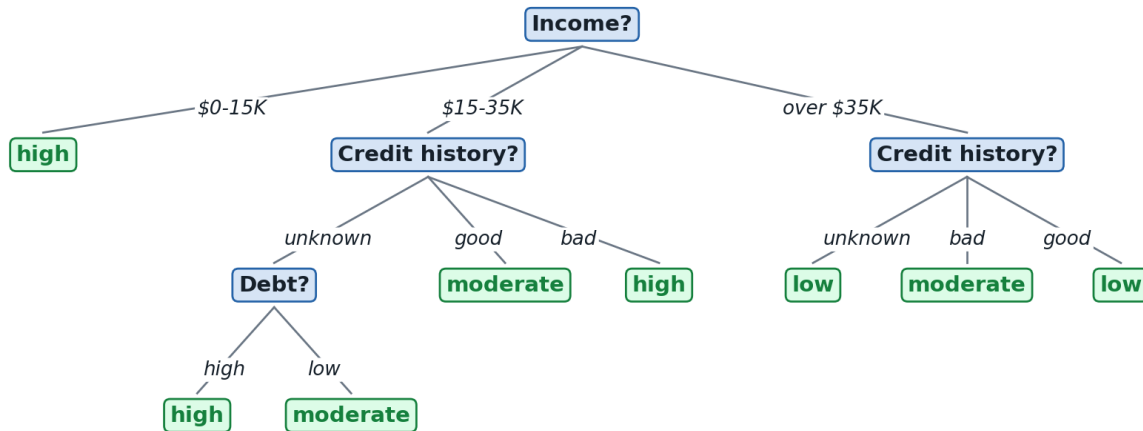
Step 4: recurse on over \$35K

6 tuples (0 high, 1 mod, 5 low), $Info = 0.650$

Attribute	$Info_A$	Gain
Credit history	unknown (0,0,2) · bad (0,1,0) · good (0,0,3) → 0	0.650
Collateral	0.459	0.191
Debt	0.541	0.109

- **Credit history splits this branch perfectly** – all three partitions are pure, so $Info_A = 0$ and the branch is finished in one step.

Final tree



Drawn by `dtfig.py`, which reruns ID3 on the table and checks the gains and the tree against this page.

Reading the tree

- 8 leaves; classifies all 14 tuples correctly.
- **Income** alone decides the \$0–15K band (all high risk).
- **Credit history** is tested twice, once under each other income band: an attribute used on one path can be reused on another.
- Only the \$15–35K, unknown-history tuples need a third test, **Debt**.

The 3-mark part: evaluating the model

- **Do not evaluate on the training set** – this tree scores 100 % there by construction, which measures nothing.
- Hold out data, or better **stratified 10-fold cross-validation** (§3.7).
- Build a **confusion matrix** on the held-out tuples, then report **accuracy**, and per class **precision**, **recall** and F_1 . With three classes, report them **per class** – a single accuracy hides which risk band is being missed.
- Mention **pruning** and a validation set, since a fully grown ID3 tree overfits.

Problem 1.4 – root node only 2+8

QUESTION AS PRINTED 2081 Bhadra · Q5 · [2+8]

State an algorithm for constructing a decision tree. Using ID3 identify the root node for the following data set.

Gender	Car Ownership	Travel Cost	Income Level	Transport Mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	High	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car

A different 10-row table from Problem 1.2, despite looking almost identical – four rows differ. **Only the root is asked for**, so stop after the gain table; do not grow the whole tree.

Step 1

classes Bus = 4, Train = 3, Car = 3, so $Info(D) = -\frac{4}{10} \log_2 \frac{4}{10} - 2\left(\frac{3}{10} \log_2 \frac{3}{10}\right) = 1.571$

Step 2: gain of each attribute

partitions written as (Bus, Train, Car)

Attribute	Partition (Bus, Train, Car)	$Info_A(D)$	Gain
Travel cost	Cheap (4,1,0) · Standard (0,2,0) pure · Expensive (0,0,3)	$\frac{5}{10}(0.722) + \frac{2}{10}(0) + \frac{3}{10}(0) = 0.361$	1.210
Car ownership	0 (2,1,0) · 1 (2,2,1) · 2 (0,0,2)	$\frac{3}{10}(0.918) + \frac{5}{10}(1.522) + \frac{2}{10}(0) = 1.036$	0.534
Income level	Low (2,0,0) pure · Medium (2,2,1) · High (0,1,2)	$\frac{2}{10}(0) + \frac{5}{10}(1.522) + \frac{3}{10}(0.918) = 1.036$	0.534
Gender	Male (3,1,1) · Female (1,2,2)	$\frac{5}{10}(1.371) + \frac{5}{10}(1.446) = 1.446$	0.125

Answer

- **Travel cost is the root node**, gain 1.210, more than double the next best.
- **Car ownership and Income level tie exactly at 0.534**. Note it – if the question had asked for the second level you would have to break that tie explicitly.
- **Why Travel cost wins**: two of its three branches are already pure, so it removes most of the uncertainty in one test. **Say that**, do not just quote the number.

M2. Gini index**Method**

$$Gini(D) = 1 - \sum_i p_i^2 \quad Gini_A(D) = \frac{|D_1|}{|D|} Gini(D_1) + \frac{|D_2|}{|D|} Gini(D_2) \quad \Delta Gini = Gini(D) - Gini_A(D)$$

- **Gini uses BINARY splits only**. For an attribute with v values you must try all $2^{v-1} - 1$ ways of splitting the values into 2 subsets and keep the best.
- Pick the largest $\Delta Gini$ (equivalently the smallest $Gini_A$).

Worked example (same 14-tuple dataset) 6

QUESTION AS PRINTED 2075 Chaitra · Q3 · [6+4]

Define Decision Tree Classifier with Gini-Index with suitable example. How can you handle overfitting in Decision Tree?

QUESTION AS PRINTED 2071 Chaitra · Q3 · [10]

What is a decision tree? Explain Gini Index with suitable example.

Neither paper prints a dataset – the “suitable example” is yours to supply. The 14-tuple *buys_computer* set below is the safe one to reuse. verified/added

- $Gini(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 1 - 0.4133 - 0.1276 = \mathbf{0.459}$
- Split **Income** as {low, medium} vs {high}:
 - $D_1 = 10$ tuples (7 Yes, 3 No): $Gini = 1 - (0.7)^2 - (0.3)^2 = 0.420$
 - $D_2 = 4$ tuples (2 Yes, 2 No): $Gini = 1 - (0.5)^2 - (0.5)^2 = 0.500$
 - $Gini_{Income}(D) = \frac{10}{14}(0.420) + \frac{4}{14}(0.500) = 0.300 + 0.143 = 0.443$
 - $\Delta Gini = 0.459 - 0.443 = \mathbf{0.016}$
- Split **Age** as {youth, senior} vs {middle_aged}:
 - $D_1 = 10$ tuples (5 Yes, 5 No): $Gini = 0.500$
 - $D_2 = 4$ tuples (4 Yes, 0 No): $Gini = \mathbf{0}$ (pure)
 - $Gini_{Age}(D) = \frac{10}{14}(0.500) + \frac{4}{14}(0) = 0.357$
 - $\Delta Gini = 0.459 - 0.357 = \mathbf{0.102} \rightarrow \text{best, so Age is chosen}$
- Same winner as information gain, which is the usual (not guaranteed) outcome.

M3. Naive Bayes classification

Method

1. Priors: $P(C_i) = |C_{i,D}|/|D|$.
2. For each attribute value in X : $P(x_k|C_i)$ from counts within class C_i .
3. $P(X|C_i) = \prod_k P(x_k|C_i)$.
4. Compare $P(X|C_i)P(C_i)$ across classes; **largest wins**.
 - **Never divide by $P(X)$** – it is identical for every class and cancels.
 - Any zero count $\rightarrow$ apply **Laplacian correction** $\frac{n_k+1}{|C_{i,D}|+k}$.
 - Lay the work out as a small table of conditional probabilities; it earns method marks even if arithmetic slips.

Problem 3.1 10

QUESTION AS PRINTED 2072 Chaitra · Q3 · [10]

Predict Class label using naive Bayesian classifier for $X = (\text{age} = \text{youth}, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit-rating} = \text{fair})$ using the following data set.

RID	Age	Income	Student	Credit-rating	Class Buy computer
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
3	Middle-age	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes
5	Senior	Low	Yes	Fair	Yes
6	Senior	Low	Yes	Excellent	No
7	Middle-age	Low	Yes	Excellent	Yes
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
10	Senior	Medium	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes
12	Middle-age	Medium	No	Excellent	Yes
13	Middle-age	High	Yes	Fair	Yes
14	Senior	Medium	No	Excellent	No

Given

14 tuples. $P(\text{Yes}) = 9/14 = 0.643$, $P(\text{No}) = 5/14 = 0.357$.

Conditional probabilities

	Yes	No
$P(\text{age} = \text{youth})$	$2/9 = 0.222$	$3/5 = 0.600$
$P(\text{income} = \text{med})$	$4/9 = 0.444$	$2/5 = 0.400$
$P(\text{student} = \text{yes})$	$6/9 = 0.667$	$1/5 = 0.200$
$P(\text{credit} = \text{fair})$	$6/9 = 0.667$	$2/5 = 0.400$

Combine

- $P(X|\text{Yes}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = \mathbf{0.0439}$
- $P(X|\text{No}) = 0.600 \times 0.400 \times 0.200 \times 0.400 = \mathbf{0.0192}$
- $P(X|\text{Yes})P(\text{Yes}) = 0.0439 \times 0.643 = \mathbf{0.028}$
- $P(X|\text{No})P(\text{No}) = 0.0192 \times 0.357 = \mathbf{0.007}$

Final answer

$0.028 > 0.007 \rightarrow \text{buys_computer} = \text{Yes}$.

- Matches Han & Kamber 3rd ed Example 8.4.

Problem 3.2 6

QUESTION AS PRINTED 2079 Bhadra · Q4 · [6]

Suppose you have a test record " $X = (\text{Home Owner} = \text{No}, \text{Material Status} = \text{Married}, \text{Income} = \$120\text{K})$ ". Your job is to classify this record using Naive Bayesian Classification. Use the following table for your calculations.

Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Given

Defaulted: **No = 7, Yes = 3**. $P(\text{No}) = 0.7$, $P(\text{Yes}) = 0.3$.

Step 1: categorical attributes

- $P(\text{HO} = \text{No} | \text{No}) = 4/7 = 0.571$ $P(\text{HO} = \text{No} | \text{Yes}) = 3/3 = 1.0$
- $P(\text{Married} | \text{No}) = 4/7 = 0.571$ $P(\text{Married} | \text{Yes}) = 0/3 = 0 \leftarrow \text{zero!}$
- Apply **Laplacian correction** (Marital Status has $k = 3$ values): $P(\text{Married} | \text{Yes}) = \frac{0+1}{3+3} = 0.167$; and $P(\text{HO} = \text{No} | \text{Yes}) = \frac{3+1}{3+2} = 0.8$.

Step 2: continuous attribute (Income), Gaussian

$$g(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Class **No**: incomes 125, 100, 70, 120, 60, 220, 75 $\rightarrow \mu = 110, \sigma = 54.54$

$$g(120) = \frac{1}{\sqrt{2\pi}(54.54)} e^{-\frac{(10)^2}{2(2975)}} = 0.00719$$

- Class **Yes**: incomes 95, 85, 90 $\rightarrow \mu = 90, \sigma = 5$

$$g(120) = \frac{1}{\sqrt{2\pi}(5)} e^{-\frac{(30)^2}{2(25)}} = 0.0798 e^{-18} \approx 1.2 \times 10^{-9}$$

- **The Yes class has a tiny spread ($\sigma = 5$) around \$90K, so \$120K is 6 standard deviations out** — effectively impossible under that model.

Step 3: combine

- $P(X|\text{No})P(\text{No}) = 0.571 \times 0.571 \times 0.00719 \times 0.7 = \mathbf{1.64 \times 10^{-3}}$
- $P(X|\text{Yes})P(\text{Yes}) = 0.8 \times 0.167 \times 1.2 \times 10^{-9} \times 0.3 \approx \mathbf{4.8 \times 10^{-11}}$

Final answer

Defaulted Borrower = No (larger by roughly 10^7).

M4. k -Nearest Neighbour**Method**

1. Compute the Euclidean distance from the query to **every** training tuple.
 2. Sort ascending, take the k smallest.
 3. **Majority class** among those k is the prediction.
- **Show the full distance table** — that is where the marks are, not the final label.
 - You may compare squared distances to save time, but write the roots in the final table.

Problem 4.1 6

QUESTION AS PRINTED 2080 Bhadra · Q2 (b) · [3+6]

When do we use classifier? You have the following information about the flower. Your job is to classify the given flower with SepalLength = 7, SepalWidth = 3.2, PetalLength = 4.7, and PetalWidth = 1.4. Use KNN algorithm for $K = 3$ with Euclidean distance matrix.

Id	Sepal_Lenth	Spal_Width	Petal_Lenth	Petal_Width	Labal
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	6	2.2	4	1	versicolor
5	6.1	2.9	4.7	1.4	versicolor
6	5.6	2.9	3.6	1.3	versicolor
7	6.7	3.1	4.4	1.4	versicolor

Column names are

the paper's own spellings.

Id	Tuple	Squared differences	d	Label
1	(5.1, 3.5, 1.4, 0.2)	$3.61 + 0.09 + 10.89 + 1.44 = 16.03$	4.004	setosa
2	(4.9, 3.0, 1.4, 0.2)	$4.41 + 0.04 + 10.89 + 1.44 = 16.78$	4.096	setosa
3	(4.7, 3.2, 1.3, 0.2)	$5.29 + 0 + 11.56 + 1.44 = 18.29$	4.277	setosa
4	(6.0, 2.2, 4.0, 1.0)	$1 + 1 + 0.49 + 0.16 = 2.65$	1.628	versicolor
5	(6.1, 2.9, 4.7, 1.4)	$0.81 + 0.09 + 0 + 0 = 0.90$	0.949	versicolor
6	(5.6, 2.9, 3.6, 1.3)	$1.96 + 0.09 + 1.21 + 0.01 = 3.27$	1.808	versicolor
7	(6.7, 3.1, 4.4, 1.4)	$0.09 + 0.01 + 0.09 + 0 = 0.19$	0.436	versicolor

Step: pick the 3 nearest

- Sorted: $d_7 = 0.436 < d_5 = 0.949 < d_4 = 1.628 < d_6 = 1.808 < d_1 = 4.004 \dots$
- The 3 nearest are **Id 7, Id 5, Id 4** → versicolor, versicolor, versicolor.

Final answer

versicolor (unanimous, 3 of 3).

- Note all 3 setosa tuples sit at distance > 4 : they have tiny petals, so they are nowhere near the query. The vote was never in doubt.

M5. Confusion matrix metrics (8 papers — highest-yield numerical in the syllabus)

DO THIS FIRST, EVERY TIME. Copy the row and column headings onto your answer sheet and mark which axis is **actual**. The IOE papers **flip the orientation between years**.

Proof: 75 Ash and 72 Ch print the *identical* four numbers 21, 6, 7, 41 — but 75 Ash has actual on the rows and 72 Ch has predicted on the rows. The result is that **sensitivity and precision swap places** (0.778 / 0.750 becomes 0.750 / 0.778). Same numbers, different answers. verified/added

Method

- Identify TP , FN , FP , TN from the orientation. Then $P = TP + FN$ (all actual positives), $N = FP + TN$ (all actual negatives).
- Accuracy = $\frac{TP+TN}{P+N}$ Error = 1–accuracy
- Sensitivity = recall = $TPR = \frac{TP}{P}$ Specificity = $TNR = \frac{TN}{N}$
- FPR = false alarm rate = $\frac{FP}{N} = 1 - \text{specificity}$ Precision = $\frac{TP}{TP+FP}$

Problem 5.1 5

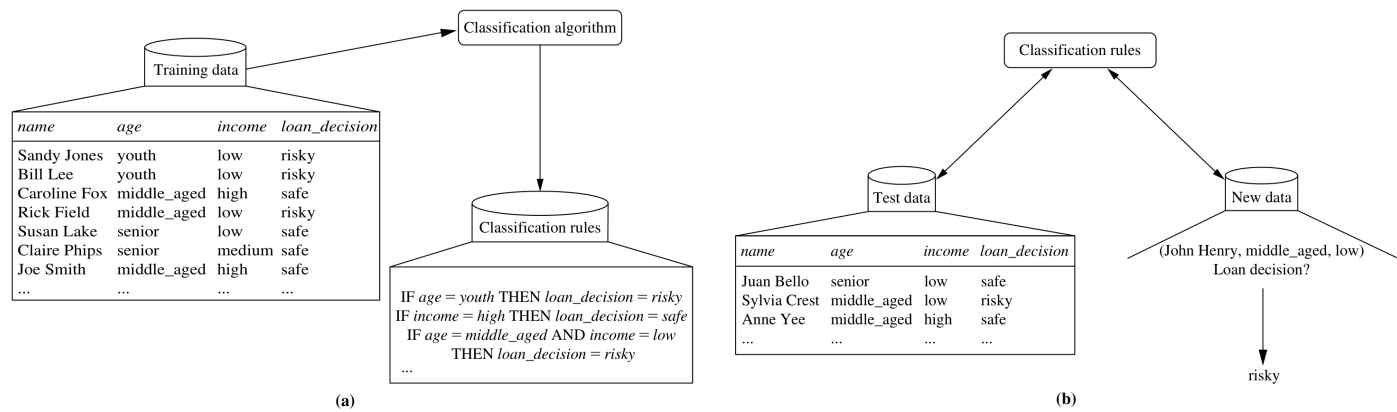
QUESTION AS PRINTED 2074 Chaitra · Q3 · [2+3+5]

Draw clear block diagram depicting different stages in classification. Explain the inverse relation between precision and recall. Given the confusion matrix, determine accuracy, sensitivity and precision of the classifier model.

Actual \ Predicted	Positive	Negative
Positive	142	40
Negative	98	720

"Positive" is the paper's own spelling.

The [2] block diagram



(a) learning: training data → algorithm → model; (b) classification: model checked on test data, then applied to new data. The precision-recall part is Ch 3 theory, §3.7. H&K Fig 8.1.

Step 1: read the four cells

rows are **Actual here**, so $TP = 142$, $FN = 40$, $FP = 98$, $TN = 720 \rightarrow P = 182$, $N = 818$, total = 1000.

Metric	Formula	Substitution	Value
Accuracy	$(TP + TN)/(P + N)$	$(142 + 720)/1000$	0.862
Sensitivity = TPR	TP/P	$142/182$	0.780
Precision	$TP/(TP + FP)$	$142/240$	0.592
Specificity	TN/N	$720/818$	0.880

Problem 5.2 4

QUESTION AS PRINTED 2078 Bhadra · Q6 · [4]

Calculate: Accuracy, TPR, FPR and Precision for the given confusion matrix for a classifier.

		Actual Class	
		Class 1	Class 2
Predicted Class	Class 1	142	40
	Class 2	98	720

Same four numbers as 5.1, but the rows are **Predicted here**.

Step 1: read the four cells

rows are **Predicted here**, so $TP = 142$, $FP = 40$, $FN = 98$, $TN = 720 \rightarrow P = 142 + 98 = 240$, $N = 40 + 720 = 760$.

Metric	Formula	Substitution	Value
Accuracy	$(TP + TN)/(P + N)$	$(142 + 720)/1000$	0.862
TPR	TP/P	$142/240$	0.592
FPR	FP/N	$40/760$	0.053
Precision	$TP/(TP + FP)$	$142/182$	0.780

- Accuracy is **unchanged from 5.1** – the diagonal is the same either way round.
- **Compare with 5.1:** sensitivity and precision have exchanged values. This is the orientation trap, live.

Problems 5.3 and 5.4 10

QUESTION AS PRINTED 2075 Ashwin · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		Predicted Class	
		Class 1	Class 2
Actual Class	Class 1	21	6
	Class 2	7	41

Calculate: Accuracy, Sensitivity, Specificity and Precision.

QUESTION AS PRINTED 2072 Chaitra · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		actual class	
		class1	class2
predicted class	class1	21	6
	class2	7	41

Calculate a. accuracy b. sensitivity c. specificity d. precision e. recall

Same four numbers, opposite orientation – that is the whole point of solving them side by side.

Both at once, so the orientation trap is visible

- **75 Ash**, rows are **Actual**: $TP = 21$, $FN = 6$, $FP = 7$, $TN = 41$; $P = 27$, $N = 48$.
- **72 Ch**, rows are **Predicted**: $TP = 21$, $FP = 6$, $FN = 7$, $TN = 41$; $P = 28$, $N = 47$.
- Both total 75.

Metric	Formula	75 Ash (rows Actual)		72 Ch (rows Predicted)	
		Subst.	Value	Subst.	Value
Accuracy	$(TP+TN)/(P+N)$	62/75	0.827	62/75	0.827
Sensitivity = recall	TP/P	21/27	0.778	21/28	0.750
Specificity	TN/N	41/48	0.854	41/47	0.872
Precision	$TP/(TP + FP)$	21/28	0.750	21/27	0.778

- **Accuracy is identical, every other metric moves.** Sensitivity and precision simply swap. Read the row/column labels before touching the arithmetic.

Problem 5.5 10

QUESTION AS PRINTED 2074 Ashwin · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		Predicted Class	
		Class 1	Class 2
Actual Class	Class 1	25	9
	Class 2	4	31

Calculate: a) Accuracy b) Sensitivity c) Specificity d) Precision

Step 1: read the four cells

rows are Actual, so $TP = 25$, $FN = 9$, $FP = 4$, $TN = 31 \rightarrow P = 34$, $N = 35$, total 69.

Metric	Formula	Substitution	Value
Accuracy	$(TP + TN)/(P + N)$	56/69	0.812
Sensitivity	TP/P	25/34	0.735
Specificity	TN/N	31/35	0.886
Precision	$TP/(TP + FP)$	25/29	0.862

Problem 5.6 3+5

QUESTION AS PRINTED 2080 Baishakh · Q3 · [3+5]

In what cases you cannot use Accuracy for performance measure, give some examples. Assume that you have the following confusion matrix. Calculate the Classification error, Sensitivity, False alarm rate Specificity.

		Actual Values	
		True	False
Predicated Values	True	1050	250
	False	150	950

Step 1: read the four cells

rows are Predicted here, so $TP = 1050$, $FP = 250$, $FN = 150$, $TN = 950 \rightarrow P = 1200$, $N = 1200$, total 2400.

Metric	Formula	Substitution	Value
Classification error	$(FP + FN)/(P + N)$	400/2400	0.167
Sensitivity	TP/P	1050/1200	0.875
False alarm rate (FPR)	FP/N	250/1200	0.208
Specificity	TN/N	950/1200	0.792

– **Check:** $FPR + specificity = 0.208 + 0.792 = 1$ ✓

Problem 5.7 3+6

QUESTION AS PRINTED 2073 Shrawan · Q4 · [1+3+6]

Explain ROC. Using the following data, calculate TPR, FPR, precision for given confusion matrix.

	A	B
A	20	5
B	10	40

Classify, A = Yes, B = No

Step 1: read the four cells

A = Yes is the positive class, rows are Actual, so $TP = 20$, $FN = 5$, $FP = 10$, $TN = 40 \rightarrow P = 25$, $N = 50$.

Metric	Formula	Substitution	Value
TPR	TP/P	20/25	0.80
FPR	FP/N	10/50	0.20
Precision	$TP/(TP + FP)$	20/30	0.667

– The ROC point for this classifier is (0.20, 0.80), comfortably above the diagonal.

Problem 5.8 10

QUESTION AS PRINTED 2070 Chaitra · Q3 · [10]

An input sequence “A A B B B A A B B” was used for classification. The Classifier ‘X’ predicted the sequences as: “A A B B B A A B B” where as the Classifier ‘Y’ predicted the sequences as: “A A A A B B A A B”. Develop the corresponding confusion matrix for the classifiers and find their corresponding.

i) Accuracy ii) Precision iii) True Positive Rate iv) False Positive Rate

Step 1: tabulate Y position by position

(take A as the positive class)

Position	1	2	3	4	5	6	7	8	9	10
Actual	A	A	B	B	B	A	A	A	B	B
Y	A	A	A	A	B	B	A	A	A	B
Category	TP	TP	FP	FP	TN	FN	TP	TP	FP	TN

Step 2: the two confusion matrices, then the metrics

- X is identical to the actual sequence: $TP = 5$, $TN = 5$, $FP = 0$, $FN = 0$.
- Y from the table above: $TP = 4$, $FN = 1$, $FP = 3$, $TN = 2$; $P = 5$, $N = 5$ (total 10 ✓).

Metric	Formula	Classifier X Subst.	Value	Classifier Y Subst.	Value
Accuracy	$(TP + TN)/(P + N)$	10/10	1.00	6/10	0.60
Precision	$TP/(TP + FP)$	5/5	1.00	4/7	0.571
TPR	TP/P	5/5	1.00	4/5	0.80
FPR	FP/N	0/5	0	3/5	0.60

- **X is a perfect classifier** – its ROC point is the top-left corner (0, 1).
- **Reading:** Y has decent recall (0.80) but poor precision (0.571) – it over-predicts A, firing 3 false alarms out of 5 actual Bs. Its ROC point (0.60, 0.80) is only just above the diagonal, so it is barely better than guessing.

Problem 5.9 4

QUESTION AS PRINTED 2079 Baishakh · Q6 · [4]

Given the following confusion matrix, determine Accuracy, Error rate, Sensitivity, Specificity, Precision, Recall of the classifier model.

$n = 165$	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

This paper labels the four cells for you, so there is no orientation trap – just read them off. $P = 105$, $N = 60$, total 165. ✓

Metric	Formula	Substitution	Value
Accuracy	$(TP + TN)/(P + N)$	$(100 + 50)/165$	0.909
Error rate	$(FP + FN)/(P + N)$	$(10 + 5)/165$	0.091
Sensitivity	TP/P	$100/105$	0.952
Specificity	TN/N	$50/60$	0.833
Precision	$TP/(TP + FP)$	$100/110$	0.909
Recall	= sensitivity	$100/105$	0.952

- **Checks:** accuracy + error rate = 1 ✓
- **Recall and sensitivity are the same thing.** The question lists both, so write both rows and say they are equal – do not assume it is a trick.
- Precision equalling accuracy here is a coincidence of the numbers, not a rule.

Problem 5.10 – the mislabelled matrix 2+6

QUESTION AS PRINTED 2073 Chaitra · Q4 · [2+6]

What is ID3 algorithm? Calculate TPR, FPR and Accuracy for given confusion matrix.

	Predicted +	Predicted –
Predicted +	100	40
Predicted –	60	300

The paper labels BOTH axes “Predicted” – reproduced exactly as printed. The row stubs should read *Actual*. **Say so in the answer, take rows as Actual, and state the assumption.** That sentence is worth more than the arithmetic.

Reading rows as Actual (the intended matrix)

$TP = 100$, $FN = 40$, $FP = 60$, $TN = 300 \rightarrow P = 140$, $N = 360$, total 500.

Metric	Formula	Substitution	Value
TPR	TP/P	$100/140$	0.714
FPR	FP/N	$60/360$	0.167
Accuracy	$(TP + TN)/(P + N)$	$400/500$	0.800

If instead the rows really are Predicted

$TP = 100$, $FP = 40$, $FN = 60$, $TN = 300 \rightarrow P = 160$, $N = 340$.

Metric	Formula	Substitution	Value
TPR	TP/P	$100/160$	0.625
FPR	FP/N	$40/340$	0.118
Accuracy	$(TP + TN)/(P + N)$	$400/500$	0.800

- **Accuracy is 0.800 either way** – the diagonal does not move when you transpose. TPR and FPR do.
- **The safe exam answer:** give the first table, and add one line noting the paper’s label defect and what the numbers would become under the other reading. Full marks either way, and it shows you understand the matrix rather than a memorised formula.

M6. ANN gradients (backpropagation)

Method

- Sigmoid and its derivative: $\sigma(z) = \frac{1}{1+e^{-z}}$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.
- Loss $E = \frac{1}{2}(o - t)^2 \rightarrow \frac{\partial E}{\partial o} = (o - t)$.
- Apply the **chain rule** backwards from the loss to the weight in question.

Problem 6.1 2+3+4

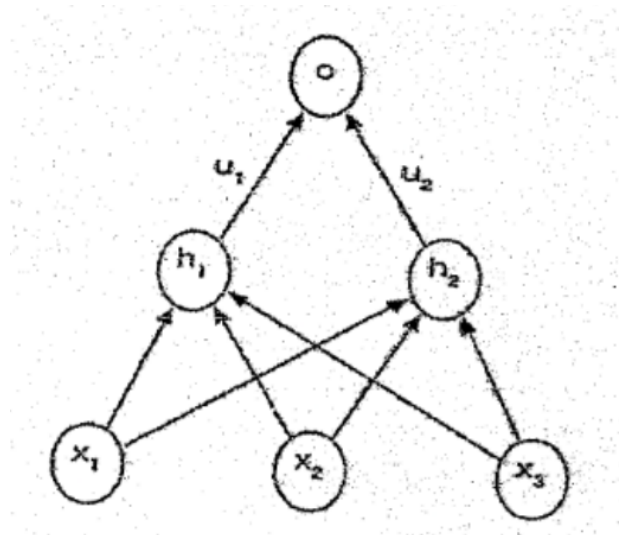
QUESTION AS PRINTED 2076 Chaitra · Q6 · [2+3+4]

Consider the multi-layer feed-forward neural network shown in the following figure. This neural network has three inputs x_1, x_2 and x_3 connected to a hidden layer consisting of two nodes h_1 and h_2 . The weight of the edge connecting x_i to h_j is w_{ji} . The two hidden nodes are connected to the output node o . The weight of the edge connecting the hidden node h_i to the output node o is u_i . The activation functions at hidden and output layers is set to sigmoid function defined as follows:

$$\sigma(\theta) = \frac{1}{1 + e^{-\theta}}$$

Using the target output t , the squared error is used as the loss function at the output node, and is defined as:

$$E(o, t) = \frac{1}{2}(o - t)^2$$



- Using the symbols given above, compute the activation at h_1 .
- Compute the gradient of the loss with respect to the output o .
- Compute the gradient of the loss with respect to the weight w_{12} .

(a) Activation at h_1

$$h_1 = \sigma(w_{11}x_1 + w_{12}x_2 + w_{13}x_3) = \frac{1}{1 + e^{-(w_{11}x_1 + w_{12}x_2 + w_{13}x_3)}}$$

(b) Gradient of the loss w.r.t. the output o

$$E(o, t) = \frac{1}{2}(o - t)^2 \Rightarrow \frac{\partial E}{\partial o} = (o - t)$$

(c) Gradient of the loss w.r.t. w_{12}

- w_{12} is the weight from x_2 into h_1 , so the chain runs $E \rightarrow o \rightarrow h_1 \rightarrow w_{12}$.
- Write $net_o = u_1h_1 + u_2h_2$ and $net_{h_1} = w_{11}x_1 + w_{12}x_2 + w_{13}x_3$.

$$\frac{\partial E}{\partial w_{12}} = \underbrace{\frac{\partial E}{\partial o}}_{(o-t)} \cdot \underbrace{\frac{\partial o}{\partial net_o}}_{o(1-o)} \cdot \underbrace{\frac{\partial net_o}{\partial h_1}}_{u_1} \cdot \underbrace{\frac{\partial h_1}{\partial net_{h_1}}}_{h_1(1-h_1)} \cdot \underbrace{\frac{\partial net_{h_1}}{\partial w_{12}}}_{x_2}$$

$$\frac{\partial E}{\partial w_{12}} = (o - t) o(1 - o) u_1 h_1(1 - h_1) x_2$$

- **Sanity checks:** the factor x_2 must appear (no input, no gradient); u_1 appears and u_2 does not, because w_{12} only influences o through h_1 .
- **Weight update:** $w_{12} \leftarrow w_{12} - l \frac{\partial E}{\partial w_{12}}$.

Ch 4 – Association Analysis – Numerical Problems & Solutions

5 methods · 14 PYQ instances · every itemset count recomputed from the printed transaction tables. **This is the heaviest numerical chapter in the syllabus.**

Methods

- **M1** Apriori: candidate generation, pruning, frequent itemsets
- **M2** Rule generation from frequent itemsets
- **M3** FP-tree construction and FP-Growth mining
- **M4** Pattern evaluation: lift and the correlation measures
- **M5** Subsequence enumeration

Convert the threshold to a COUNT before you start. “min support 50%” on 6 transactions means a count of 3. Nearly every lost mark in this chapter comes from mixing percentages and counts half way through. Write $\text{min_sup_count} = N$ at the top and use only that.

Also: **support is over ALL transactions; confidence is over the antecedent's transactions only.**

M1. Apriori

Method

1. Scan once, count items $\rightarrow$ keep those $\geq \text{min_sup_count} \rightarrow L_1$.
 2. **Join** L_{k-1} with itself: combine 2 itemsets only if their first $k-2$ items match.
 3. **Prune**: discard a candidate if *any* of its $(k-1)$ -subsets is missing from L_{k-1} .
 4. **Count** survivors against D , keep those meeting the threshold $\rightarrow L_k$.
 5. Stop when L_k is empty.
- **Show the prune step explicitly** – it is where the method marks are.

Problem 1.1 2+7

QUESTION AS PRINTED 2080 Bhadra · Q3 (b) · [2+7]

What is Association Analysis? Explain with different use cases? Use the Apriori algorithm to find the frequent itemsets. Assume minimum support count is 4 and confidence is 80%.

TID	Items
T100	F, A, C, D, G, I, M, P, N
T101	A, B, C, D, F, L, M, O, P
T102	B, F, H, V, J, O, P
T103	B, C, K, S, A, V
T104	L, A, F, C, E, P, M, N, V
T105	I, B, A, P, S, M
T106	F, A, C, I, B, A, P

T106 genuinely prints “F, A, C, I, B, A, P” – **A is repeated; a transaction is a set**, so treat it as {F,A,C,I,B,P}.

Step 1: $C_1 \rightarrow L_1$

count all 18 distinct items over the 7 transactions, min count = 4

It.	Cnt	Keep	It.	Cnt	Keep	It.	Cnt	Keep	It.	Cnt	Keep
A	6	✓	F	5	✓	K	1	×	O	2	×
B	5	✓	G	1	×	L	2	×	P	6	✓
C	5	✓	H	1	×	M	4	✓	S	2	×
D	2	×	I	3	×	N	2	×	V	3	×
E	1	×	J	1	×						

$L_1 = \{\mathbf{A(6)}, \mathbf{B(5)}, \mathbf{C(5)}, \mathbf{F(5)}, \mathbf{M(4)}, \mathbf{P(6)}\}$ – 6 items survive of 18.

Step 2: $C_2 \rightarrow L_2$

all $\binom{6}{2} = 15$ pairs from L_1 , counted against D

Pair	Cnt	Keep	Pair	Cnt	Keep	Pair	Cnt	Keep	Pair	Cnt	Keep
AB	4	✓	AP	5	✓	BP	4	✓	CP	4	✓
AC	5	✓	BC	3	×	CF	4	✓	FM	3	×
AF	4	✓	BF	3	×	CM	3	×	FP	5	✓
AM	4	✓	BM	2	×				MP	4	✓

$L_2 = \{AB(4), AC(5), AF(4), AM(4), AP(5), BP(4), CF(4), CP(4), FP(5), MP(4)\}$

Step 3: $C_3 \rightarrow L_3$

join L_2 pairs sharing their first item, then **prune before counting**

Cand.	Subset test (all 2-subsets must be in L_2)	Transactions	Count	Keep
ABC	$BC \notin L_2$	pruned, never counted	—	×
ABF	$BF \notin L_2$	pruned, never counted	—	×
ABM	$BM \notin L_2$	pruned, never counted	—	×
ABP	$AB \checkmark AP \checkmark BP \checkmark \rightarrow$ survives pruning	T101, T105, T106	3	×
ACF	$AC \checkmark AF \checkmark CF \checkmark$	T100, T101, T104, T106	4	✓
ACM	$CM \notin L_2$	pruned, never counted	—	×
ACP	$AC \checkmark AP \checkmark CP \checkmark$	T100, T101, T104, T106	4	✓
AFM	$FM \notin L_2$	pruned, never counted	—	×
AFP	$AF \checkmark AP \checkmark FP \checkmark$	T100, T101, T104, T106	4	✓
AMP	$AM \checkmark AP \checkmark MP \checkmark$	T100, T101, T104, T105	4	✓
CFP	$CF \checkmark CP \checkmark FP \checkmark$	T100, T101, T104, T106	4	✓

$L_3 = \{ACF(4), ACP(4), AFP(4), AMP(4), CFP(4)\}$

- **Five candidates died at the prune step without a single database scan** — that saving is the whole point of the Apriori property, so show the column.
- ABP is the instructive one: it *passes* pruning and still fails on the count. **Pruning is sufficient to reject, never sufficient to accept.**

Step 4: $C_4 \rightarrow L_4$

join L_3 triples sharing their first two items

Cand.	Joined from	Subset test	Count	Keep
ACFP	ACF + ACP	$ACF \checkmark ACP \checkmark AFP \checkmark CFP \checkmark$	4 (T100, T101, T104, T106)	✓

$L_4 = \{ACFP(4)\}$; no two 4-itemsets remain to join $\rightarrow C_5 = \emptyset \rightarrow$ **stop**.

Frequent itemsets

Maximal frequent itemset = **$\{A, C, F, P\}$ with support $4/7 = 57\%$.**

- **Check:** all $2^4 - 2 = 14$ proper subsets of ACFP do appear in $L_1 - L_3$ above. ✓

Step 5: the rules (the question gives confidence = 80 %, so this half is asked)

- From the maximal itemset $l = \{A, C, F, P\}$, $\sigma(l) = 4$. For every non-empty proper subset s , output $s \Rightarrow (l - s)$ with

$$conf = \frac{\sigma(l)}{\sigma(s)} = \frac{4}{\sigma(s)}.$$

- $2^4 - 2 = 14$ candidate rules. $conf \geq 0.80$ needs $\sigma(s) \leq 5$, so read the answer straight off the antecedent's count.

Rule	$\sigma(s)$	Confidence
$C \Rightarrow AFP$	5	$4/5 = 80\% \checkmark$
$F \Rightarrow ACP$	5	$4/5 = 80\% \checkmark$
$AC \Rightarrow FP$	5	$4/5 = 80\% \checkmark$
$AP \Rightarrow CF$	5	$4/5 = 80\% \checkmark$
$FP \Rightarrow AC$	5	$4/5 = 80\% \checkmark$
$AF \Rightarrow CP$	4	$4/4 = 100\% \checkmark$
$CF \Rightarrow AP$	4	$4/4 = 100\% \checkmark$
$CP \Rightarrow AF$	4	$4/4 = 100\% \checkmark$
$ACF \Rightarrow P$	4	$4/4 = 100\% \checkmark$
$ACP \Rightarrow F$	4	$4/4 = 100\% \checkmark$
$AFP \Rightarrow C$	4	$4/4 = 100\% \checkmark$
$CFP \Rightarrow A$	4	$4/4 = 100\% \checkmark$
$A \Rightarrow CFP$	6	$4/6 = 66.7\% \times$
$P \Rightarrow ACF$	6	$4/6 = 66.7\% \times$

Reading the table

- **12 of the 14 rules are strong** at min_conf = 80%.
- The only two failures are the ones with A or P alone on the left. Both occur in 6 of the 7 transactions, so they are **too common to predict anything**: a frequent antecedent divides the confidence down.
- **The general lesson**: confidence falls as the antecedent gets more frequent, and rises as it gets more specific. Longer left-hand sides almost always score higher.
- Seven rules here sit at **100%**. By §4.1 those are the *least* interesting of the twelve, because $\{A, F\}$ never appears without C and P anywhere in this database → nothing actionable. **Say so if the question asks you to comment**, do not just report the highest number.

Problem 1.2 5

QUESTION AS PRINTED 2080 Baishakh · Q5 · [4+5]

Generally, we will be more interested in associated rules with high confidence. However, often we will not be interested in association rules that have a confidence of 100%. Why? Then specifically explain why association rules with 99% confidence may be interesting (i.e., what might they indicate)? Identify the candidate and large item sets of the following transaction table using Apriori algorithm with minimum support 2.

TID	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

Step 1: $C_1 \rightarrow L_1$

min count = 2

Item	In	Count	Keep
A	10, 30	2	$\checkmark$
B	20, 30, 40	3	$\checkmark$
C	10, 20, 30	3	$\checkmark$
D	10	1	$\times$
E	20, 30, 40	3	$\checkmark$

$L_1 = \{A(2), B(3), C(3), E(3)\}$

Step 2: $C_2 \rightarrow L_2$

all $\binom{4}{2} = 6$ pairs from L_1

Pair	In	Count	Keep
AB	30	1	$\times$
AC	10, 30	2	$\checkmark$
AE	30	1	$\times$
BC	20, 30	2	$\checkmark$
BE	20, 30, 40	3	$\checkmark$
CE	20, 30	2	$\checkmark$

$L_2 = \{AC(2), BC(2), BE(3), CE(2)\}$

Step 3: $C_3 \rightarrow L_3$

join L_2 on a shared first item, prune, then count

Joined from	Candidate	Subset test	In	Count	Keep
BC + BE	BCE	BC $\checkmark$ BE $\checkmark$ CE $\checkmark$ → survives	20, 30	2	$\checkmark$
AC + —	none	AC is the only pair starting with A, so it has nothing to join with	—	—	—

$L_3 = \{BCE(2)\}$; C_4 needs two 3-itemsets sharing a prefix and there is only one → **stop**.

Final answer

Frequent itemsets: A, B, C, E, AC, BC, BE, CE, BCE.

Problem 1.3 10

QUESTION AS PRINTED 2070 Chaitra · Q4 · [10]

Explain Apriori algorithm. Use Apriori to generate frequent item sets with support of 50% for the following transaction database.

TID	Items
1	ACD
2	BD
3	ABCE
4	BDF

4 transactions, 50% $\rightarrow$ min count = 2.

- L_1 : A(2), B(3), C(2), D(3) (E:1, F:1 dropped)
- C_2 : AB(1) $\times$ AC(2) $\checkmark$ AD(1) $\times$ BC(1) $\times$ BD(2) $\checkmark$ CD(1) $\times$
- $L_2 = \{\mathbf{AC(2)}, \mathbf{BD(2)}\}$
- C_3 : AC and BD share no common prefix $\rightarrow$ **no candidates**. Stop.

Final answer

Frequent itemsets: A, B, C, D, AC, BD.

Problems 1.4 and 1.5 5+3 8

QUESTION AS PRINTED 2076 Chaitra · Q7 · [5+3]

Consider the transaction data shown in the following table from a fast food restaurant.

Meal Item	List of Item IDs
Order:1	M1, M2, M5
Order:2	M2, M4
Order:3	M2, M3
Order:4	M1, M2, M4
Order:5	M1, M3
Order:6	M2, M3
Order:7	M1, M3
Order:8	M1, M2, M3, M5
Order:9	M1, M2, M3

There are 9 distinct transactions (Order: 1 – Order: 9) and each transaction involves between 2 and 4 meal items. There are a total of 5 meal items that are involved in the transactions. For simplicity, the meal items have been assigned short names (M1–M5). Assume that the minimum support is 2/9 and the minimum confidence is 7/9.

- Apply the Apriori algorithm to the dataset of transactions and identify all frequent k -itemsets.
- Find all strong association rules of the form: $X \wedge Y \rightarrow Z$ and note their confidence values.

QUESTION AS PRINTED 2074 Ashwin · Q5 · [8]

Identify the candidate, frequent item sets and association rules for the following transaction data using Apriori algorithm.

TID	ITEMS
1	M1, M2, M5
2	M2, M4
3	M2, M3
4	M1, M2, M4
5	M1, M3
6	M2, M3
7	M1, M3
8	M1, M2, M3, M5
9	M1, M2, M3

Take minimum support = 20%, minimum confidence 80%

Identical 9-transaction table, and both thresholds work out to a count of 2, so the frequent itemsets are the same; only the rule filter differs.

Frequent itemsets

(9 transactions, min count 2)

Step 1: $C_1 \rightarrow L_1$

Item	In	Count	Keep
M1	T1,4,5,7,8,9	6	✓
M2	T1,2,3,4,6,8,9	7	✓
M3	T3,5,6,7,8,9	6	✓
M4	T2,4	2	✓
M5	T1,8	2	✓

Nothing is dropped — all 5 items clear a count of 2.

Step 2: $C_2 \rightarrow L_2$

all $\binom{5}{2} = 10$ pairs

Pair	In	Count	Keep
M1M2	T1, T4, T8, T9	4	✓
M1M3	T5, T7, T8, T9	4	✓
M1M4	T4	1	×
M1M5	T1, T8	2	✓
M2M3	T3, T6, T8, T9	4	✓
M2M4	T2, T4	2	✓
M2M5	T1, T8	2	✓
M3M4	—	0	×
M3M5	T8	1	×
M4M5	—	0	×

$L_2 = \{\text{M1M2(4), M1M3(4), M1M5(2), M2M3(4), M2M4(2), M2M5(2)}\}$

Step 3: $C_3 \rightarrow L_3$

join two L_2 pairs **only when both are in L_2 and they share their first item**

Joined from	Candidate	Subset test	In	Count	Keep
M1M2 + M1M3	M1M2M3	M1M2 ✓ M1M3 ✓ M2M3 ✓	T8, T9	2	✓
M1M2 + M1M5	M1M2M5	M1M2 ✓ M1M5 ✓ M2M5 ✓	T1, T8	2	✓
M1M3 + M1M5	M1M3M5	M3M5 $\notin L_2 \rightarrow$ pruned	—	—	×
M2M3 + M2M4	M2M3M4	M3M4 $\notin L_2 \rightarrow$ pruned	—	—	×
M2M3 + M2M5	M2M3M5	M3M5 $\notin L_2 \rightarrow$ pruned	—	—	×
M2M4 + M2M5	M2M4M5	M4M5 $\notin L_2 \rightarrow$ pruned	—	—	×

$L_3 = \{\text{M1M2M3(2), M1M2M5(2)}\}$

- **Note what is not in the table:** M1M2M4 never appears, because generating it would need M1M4, and M1M4 failed at step 2. **A candidate is only ever built from two surviving parents** — it is not built and then pruned.

Step 4: $C_4 \rightarrow L_4$

Joined from	Candidate	Subset test	Count	Keep
M1M2M3 + M1M2M5	M1M2M3M5	M1M3M5 $\notin L_3 \rightarrow$ pruned	—	×

$L_4 = \emptyset \rightarrow$ **stop.**

The one idea that decides every rule below

- Every rule out of an itemset l has the *same* numerator: $\text{conf}(s \Rightarrow l - s) = \sigma(l)/\sigma(s)$. **Only the antecedent's count moves.**
- So a rule is strong exactly when **the antecedent is nearly as rare as the whole itemset** $\rightarrow$ when the antecedent hardly ever appears without the consequent.
- Here $\sigma(\text{M4}) = \sigma(\text{M5}) = 2$, and both *always* appear with their partners. **Expect every strong rule to have M4 or M5 on the left**, and check that prediction against the tables. Nothing else can win, because M1, M2 and M3 each occur 6–7 times while no itemset containing them has support above 4.

1.4 Rules of the form $X \wedge Y \Rightarrow Z$, $\min \text{conf} = 7/9 = 0.778$

- That form means a **2-item antecedent and a 1-item consequent**, so only the **3-itemsets** in L_3 can produce one. Two of them, three rules each $\rightarrow$ **6 candidates, and they are all of them.**

From	Rule	$\sigma(XY)$	Conf = $2/\sigma(XY)$
M1M2M3 $\sigma = 2$	M1 $\wedge$ M2 $\Rightarrow$ M3	4	0.50 $\times$
	M1 $\wedge$ M3 $\Rightarrow$ M2	4	0.50 $\times$
	M2 $\wedge$ M3 $\Rightarrow$ M1	4	0.50 $\times$
M1M2M5 $\sigma = 2$	M1 $\wedge$ M2 $\Rightarrow$ M5	4	0.50 $\times$
	M1$\wedge$M5$\Rightarrow$M2	2	1.00 $\checkmark$
	M2$\wedge$M5$\Rightarrow$M1	2	1.00 $\checkmark$

Answer to 1.4

- **2 strong rules, both at 100 % confidence:**
M1 $\wedge$ M5 $\Rightarrow$ M2 and M2 $\wedge$ M5 $\Rightarrow$ M1.
- Every rule out of M1M2M3 fails, and all three fail *identically* at 0.50, because all three of its 2-subsets happen to have $\sigma = 4$.
- The two winners are exactly the ones predicted above: **M5 is in the antecedent**, and M5 occurs only in T1 and T8, both of which also hold M1 and M2.

1.5 All association rules, min conf = 0.80

- Now every frequent itemset with $|I| \geq 2$ contributes: six 2-itemsets (2 rules each) and two 3-itemsets ($2^3 - 2 = 6$ rules each) $\rightarrow 12 + 12 = 24$ **candidate rules**. All 24 are listed, none is skipped.

From	Rule	$\sigma(s)$	Conf	From	Rule	$\sigma(s)$	Conf
M1M2 (4)	M1 $\Rightarrow$ M2	6	0.67 $\times$	M2M5 (2)	M2 $\Rightarrow$ M5	7	0.29 $\times$
	M2 $\Rightarrow$ M1	7	0.57 $\times$		M5$\Rightarrow$M2	2	1.00 $\checkmark$
M1M3 (4)	M1 $\Rightarrow$ M3	6	0.67 $\times$	M1M2M3 (2)	M1 $\Rightarrow$ M2M3	6	0.33 $\times$
	M3 $\Rightarrow$ M1	6	0.67 $\times$		M2 $\Rightarrow$ M1M3	7	0.29 $\times$
M1M5 (2)	M1 $\Rightarrow$ M5	6	0.33 $\times$		M3 $\Rightarrow$ M1M2	6	0.33 $\times$
	M5$\Rightarrow$M1	2	1.00 $\checkmark$		M1 $\wedge$ M2 $\Rightarrow$ M3	4	0.50 $\times$
M2M3 (4)	M2 $\Rightarrow$ M3	7	0.57 $\times$		M1 $\wedge$ M3 $\Rightarrow$ M2	4	0.50 $\times$
	M3 $\Rightarrow$ M2	6	0.67 $\times$	M1M2M5 (2)	M2 $\wedge$ M3 $\Rightarrow$ M1	4	0.50 $\times$
M2M4 (2)	M2 $\Rightarrow$ M4	7	0.29 $\times$		M1 $\Rightarrow$ M2M5	6	0.33 $\times$
	M4$\Rightarrow$M2	2	1.00 $\checkmark$		M2 $\Rightarrow$ M1M5	7	0.29 $\times$
					M5$\Rightarrow$M1$\wedge$M2	2	1.00 $\checkmark$
					M1 $\wedge$ M2 $\Rightarrow$ M5	4	0.50 $\times$
					M1$\wedge$M5$\Rightarrow$M2	2	1.00 $\checkmark$
					M2$\wedge$M5$\Rightarrow$M1	2	1.00 $\checkmark$

Answer to 1.5

- **6 strong rules, every one of them at 100 % confidence:**
M5 $\Rightarrow$ M1 M5 $\Rightarrow$ M2 M4 $\Rightarrow$ M2 M5 $\Rightarrow$ M1 $\wedge$ M2 M1 $\wedge$ M5 $\Rightarrow$ M2 M2 $\wedge$ M5 $\Rightarrow$ M1
- The other 18 fail, and **the prediction held**: every survivor has M4 or M5 on the left. M1, M2 and M3 are too frequent to be confident antecedents here, the best any of them manages is 0.67.
- **Cross-check with 1.4**: the two rules found there must reappear here, since 1.4's threshold (0.778) is the tighter of the two. They do, on the last two rows. $\checkmark$
- **Worth a closing sentence**: all six sit at 100 %, which by §4.1 makes them *reliable but not surprising* $\rightarrow$ M4 and M5 are simply never ordered alone. Confidence alone cannot see that; lift would.

Problem 1.6 8

QUESTION AS PRINTED 2079 Bhadra · Q5 · [8]

Derive association rule for the following market basket transactions.

Minimum support = 50% Minimum confidence = 80%

Transaction ID	Item Set
1	A,B
2	A,D
3	A,C
4	B,E
5	B,D,E
6	A,E,C

This one has a surprise. 6 transactions, 50 % $\rightarrow$ min count = 3.

Step 1: $C_1 \rightarrow L_1$

Item	In	Count	Keep
A	1, 2, 3, 6	4	✓
B	1, 4, 5	3	✓
C	3, 6	2	×
D	2, 5	2	×
E	4, 5, 6	3	✓

$L_1 = \{A(4), B(3), E(3)\}$

Final answer

- **No frequent 2-itemset exists, so no association rule can be derived at 50 % support.** Say this explicitly – it is the correct answer, not a dead end.
- And it holds **for every confidence threshold, including 0 %**: a rule needs a 2-itemset to split, and there is none. **Lowering min_conf cannot rescue it, only lowering min_sup can** (§4.2).
- **Show you understand why**: the data is spread thin, every pair occurs at most twice.
- If min support is relaxed to 33.3 % (count 2), the picture changes:

New L_2	Count	Rule	Confidence	Strong at 80 %?
AC	2	$A \Rightarrow C$	$2/4 = 0.50$	×
		$C \Rightarrow A$	$2/2 = 1.00$	✓
BE	2	$B \Rightarrow E$	$2/3 = 0.67$	×
		$E \Rightarrow B$	$2/3 = 0.67$	×

One rule survives: $C \Rightarrow A$ at 100 %. Worth adding as a closing remark. verified/added

Problem 1.7 2+8

QUESTION AS PRINTED 2082 Bhadra · Q4 · [2+8]

Explain the term support and confidence in association analysis. Derive association rule for following transaction considering minimum support = 60% and confidence = 80% using Apriori Algorithm.

TID	Transaction
T1	A, B, C, D, E, F
T2	B, C, D, E, F, G
T3	A, D, E, H
T4	A, D, F, I, J
T5	B, D, E, K

5 transactions, 60 % $\rightarrow$ **min count = 3**.

Step 1: $C_1 \rightarrow L_1$

11 distinct items

Item	In	Cnt	Keep
A	T1, T3, T4	3	✓
B	T1, T2, T5	3	✓
C	T1, T2	2	×
D	T1–T5	5	✓
E	T1, T2, T3, T5	4	✓
F	T1, T2, T4	3	✓
G	T2	1	×
H	T3	1	×
I	T4	1	×
J	T4	1	×
K	T5	1	×

$L_1 = \{A(3), B(3), D(5), E(4), F(3)\}$

Step 2: $C_2 \rightarrow L_2$

all $\binom{5}{2} = 10$ pairs

Pair	In	Cnt	Keep
AB	T1	1	×
AD	T1, T3, T4	3	✓
AE	T1, T3	2	×
AF	T1, T4	2	×
BD	T1, T2, T5	3	✓
BE	T1, T2, T5	3	✓
BF	T1, T2	2	×
DE	T1, T2, T3, T5	4	✓
DF	T1, T2, T4	3	✓
EF	T1, T2	2	×

$L_2 = \{AD(3), BD(3), BE(3), DE(4), DF(3)\}$

Step 3: $C_3 \rightarrow L_3$

Joined from	Cand.	Subset test	In	Cnt	Keep
BD + BE	BDE	BD ✓ BE ✓ DE ✓	T1, T2, T5	3	✓
AD + –	none	AD is the only pair starting with A	–	–	–
DE + DF	DEF	EF $\notin L_2 \rightarrow$ pruned	–	–	×

$L_3 = \{\mathbf{BDE(3)}\}$; $C_4 = \emptyset \rightarrow$ **stop**.

Step 4: the rules

$5 \times 2 + (2^3 - 2) = 16$ candidates, min conf 80 %

From	Rule	$\sigma(s)$	Conf	From	Rule	$\sigma(s)$	Conf
AD (3)	A $\Rightarrow$ D	3	1.00 ✓	DF (3)	D $\Rightarrow$ F	5	0.60 ×
	D $\Rightarrow$ A	5	0.60 ×		F $\Rightarrow$ D	3	1.00 ✓
BD (3)	B $\Rightarrow$ D	3	1.00 ✓	BDE (3)	B $\Rightarrow$ DE	3	1.00 ✓
	D $\Rightarrow$ B	5	0.60 ×		D $\Rightarrow$ BE	5	0.60 ×
BE (3)	B $\Rightarrow$ E	3	1.00 ✓		E $\Rightarrow$ BD	4	0.75 ×
	E $\Rightarrow$ B	4	0.75 ×		BD $\Rightarrow$ E	3	1.00 ✓
DE (4)	D $\Rightarrow$ E	5	0.80 ✓		BE $\Rightarrow$ D	3	1.00 ✓
	E $\Rightarrow$ D	4	1.00 ✓		DE $\Rightarrow$ B	4	0.75 ×

- **9 strong rules of 16.** $D \Rightarrow E$ scrapes in at exactly 0.80 – **$\geq$ the threshold counts as passing**, so keep it.
- Every failure has D or E alone on the left. D is in *all five* transactions, so it predicts nothing (§4.1); it is the consequent of four winning rules instead.

Problem 1.8 2+6

QUESTION AS PRINTED 2081 Bhadra · Q7 · [2+6]

For the transaction given below, consider Confidence = 65% and minimum support = 50%. Identify frequent itemsets with possible association rules using Apriori Algorithm.

TID	Items
1	MILK, BREAD, CAKE
2	BUTTER, BREAD, EGG
3	MILK, BUTTER, BREAD, EGG
4	BUTTER, EGG

4 transactions, 50 % $\rightarrow$ **min count = 2**. **Note the unusually low confidence bar**, 65 %, which is why almost everything survives.

Step 1: $C_1 \rightarrow L_1$

Item	In	Cnt	Keep
MILK	1, 3	2	✓
BREAD	1, 2, 3	3	✓
CAKE	1	1	×
BUTTER	2, 3, 4	3	✓
EGG	2, 3, 4	3	✓

$L_1 = \{\mathbf{MILK(2)}, \mathbf{BREAD(3)}, \mathbf{BUTTER(3)}, \mathbf{EGG(3)}\}$

Step 2: $C_2 \rightarrow L_2$

6 pairs

Pair	In	Cnt	Keep
BREAD, BUTTER	2, 3	2	✓
BREAD, EGG	2, 3	2	✓
BREAD, MILK	1, 3	2	✓
BUTTER, EGG	2, 3, 4	3	✓
BUTTER, MILK	3	1	×
EGG, MILK	3	1	×

Step 3: $C_3 \rightarrow L_3$

Joined from	Candidate	Subset test	Cnt	Keep
{BR,BU} + {BR,EG}	BREAD, BUTTER, EGG	all three 2-subsets in L_2 ✓	2	✓
{BR,BU} + {BR,MI}	BREAD, BUTTER, MILK	{BU,MI} $\notin L_2 \rightarrow$ pruned	–	×
{BR,EG} + {BR,MI}	BREAD, EGG, MILK	{EG,MI} $\notin L_2 \rightarrow$ pruned	–	×

$L_3 = \{\{\mathbf{BREAD, BUTTER, EGG}\}(2)\}$, $C_4 = \emptyset \rightarrow$ **stop**.

Step 4: the rules

$4 \times 2 + 6 = 14$ candidates, min conf 65 %

Rule	$\sigma(s)$	Conf	Rule	$\sigma(s)$	Conf
BREAD $\Rightarrow$ BUTTER	3	0.667 ✓	BUTTER $\Rightarrow$ EGG	3	1.00 ✓
BUTTER $\Rightarrow$ BREAD	3	0.667 ✓	EGG $\Rightarrow$ BUTTER	3	1.00 ✓
BREAD $\Rightarrow$ EGG	3	0.667 ✓	BREAD $\Rightarrow$ BU,EG	3	0.667 ✓
EGG $\Rightarrow$ BREAD	3	0.667 ✓	BUTTER $\Rightarrow$ BR,EG	3	0.667 ✓
BREAD $\Rightarrow$ MILK	3	0.667 ✓	EGG $\Rightarrow$ BR,BU	3	0.667 ✓
MILK $\Rightarrow$ BREAD	2	1.00 ✓	BR,BU $\Rightarrow$ EGG	2	1.00 ✓
			BR,EG $\Rightarrow$ BUTTER	2	1.00 ✓
			BU,EG $\Rightarrow$ BREAD	3	0.667 ✓

- **All 14 rules are strong.** That is the point of the 65 % threshold: the lowest confidence anywhere in this database is $2/3 = 0.667$, which clears it.
- **Say so explicitly rather than listing 14 rules and stopping** – “every candidate rule passes, because min_conf sits below the minimum achievable confidence here” is the observation worth a mark. Raise the bar to 70 % and only the five 100 % rules survive.

Problem 1.9 3+7

QUESTION AS PRINTED 2073 Chaitra · Q5 · [3+7]

Explain Apriori algorithm in market basket analysis? Derive association rule from the following market basket transactions with 50% of minimum support and confidence respectively.

Transaction	Itemsets
1	A, B, C
2	A, C
3	A, D
4	B, E, F

“50% of minimum support and confidence respectively” means **both** thresholds are 50%: min count = 2, min conf = 0.50.

Step 1: $C_1 \rightarrow L_1$

Item	In	Cnt	Keep
A	1, 2, 3	3	✓
B	1, 4	2	✓
C	1, 2	2	✓
D	3	1	×
E	4	1	×
F	4	1	×

$L_1 = \{A(3), B(2), C(2)\}$

Step 3: the rules

only $\{A, C\}$ can produce any, so $2^2 - 2 = 2$ candidates

Rule	$\sigma(s)$	Confidence	Strong at 50 %?
$A \Rightarrow C$	3	$2/3 = 0.667$	✓
$C \Rightarrow A$	2	$2/2 = 1.000$	✓

- **Both are strong:** $A \Rightarrow C$ (67 %) and $C \Rightarrow A$ (100 %).
- **Transaction 4 contributes nothing.** $\{B, E, F\}$ shares no item pattern with the rest, which is why B survives as a 1-itemset but never appears in a rule. Worth a sentence.

Problem 1.10 5+3

QUESTION AS PRINTED 2079 Baishakh · Q5 · [5+3]

a) Write Apriori algorithm and using the algorithm find all the frequent itemset for the following database. (min_sup = 20%).

TID	A1	A2	A3	A4	A5	A6	A7	A8	A9
T2	0	1	0	1	0	0	0	1	0
T3	0	0	0	1	1	0	1	0	0
T4	0	0	1	0	0	0	0	0	0
T5	0	0	0	0	1	1	1	0	0
T6	0	1	1	1	0	0	0	0	0
T7	0	1	0	0	0	1	1	0	1
T8	0	0	0	0	1	0	0	0	0

b) Is it possible to generate any rules out of the frequent items, considering any value of confidence threshold?

The table really does start at T2, has seven rows, and its 8th column really is headed A8 (the scan clips the 8). 20 % of 7 transactions = 1.4, and a count cannot be fractional, so **min count = 2**. **Column A1 is 0 in every row** — it can never be frequent, so drop it at once and say why.

(a) Step 1: $C_1 \rightarrow L_1$

Step 2: $C_2 \rightarrow L_2$

$\binom{6}{2} = 15$ pairs

Item	In	Cnt	Keep	Pair	Cnt	Keep	Pair	Cnt	Keep
A1	— (all zero)	0	×	A2,A3	1	×	A4,A5	1	×
A2	T2, T6, T7	3	✓	A2,A4	2	✓	A4,A6	0	×
A3	T4, T6	2	✓	A2,A5	0	×	A4,A7	1	×
A4	T2, T3, T6	3	✓	A2,A6	1	×	A5,A6	1	×
A5	T3, T5, T8	3	✓	A2,A7	1	×	A5,A7	2	✓
A6	T5, T7	2	✓	A3,A4	1	×	A6,A7	2	✓
A7	T3, T5, T7	3	✓	A3,A5	0	×			
A8	T2	1	×	A3,A6	0	×			
A9	T7	1	×	A3,A7	0	×			

Step 3: C_3

- $L_2 = \{A2A4(2), A5A7(2), A6A7(2)\}$.
- To build a 3-itemset two pairs must **share their first item**. A2A4, A5A7 and A6A7 all start with a different item, so **no candidate can be formed at all** → $C_3 = \emptyset$ → stop.
- **Frequent itemsets:** A2, A3, A4, A5, A6, A7, {A2,A4}, {A5,A7}, {A6,A7}.

(b) Can any rule be generated, at any confidence threshold?

- **Yes**. The question is a trap only if you stop at L_1 : a rule needs a frequent itemset of size ≥ 2 to split into antecedent and consequent, and here **three 2-itemsets survive**, so rules exist.
- Each 2-itemset yields $2^2 - 2 = 2$ rules, so **6 rules in total**:

Rule	$\sigma(s)$	Confidence	Comment
A2 $\Rightarrow$ A4	3	$2/3 = 0.667$	
A4 $\Rightarrow$ A2	3	$2/3 = 0.667$	
A5 $\Rightarrow$ A7	3	$2/3 = 0.667$	
A7 $\Rightarrow$ A5	3	$2/3 = 0.667$	
A6 $\Rightarrow$ A7	2	$2/2 = 1.000$	A6 never occurs without A7
A7 $\Rightarrow$ A6	3	$2/3 = 0.667$	

- So the honest answer is: **rules exist for any threshold up to 0.667**, and above that only **A6 $\Rightarrow$ A7** survives. Above 1.0 nothing does.
- **The general principle** (§4.2): rules exist iff $L_2 \neq \emptyset$. **Check L_2 , not the confidence value** — if L_2 were empty no threshold, not even 0 %, could produce a rule. Compare Problem 1.6, where exactly that happens.

M2. Rule generation from frequent itemsets

Method

- For each frequent itemset l with $|l| \geq 2$, take every non-empty proper subset s :

$$s \Rightarrow (l - s) \quad \text{with} \quad \text{conf} = \frac{\text{sup_count}(l)}{\text{sup_count}(s)}$$

- Keep it if $\text{conf} \geq \text{min_conf}$. A k -itemset gives $2^k - 2$ candidate rules.
- The support of every rule from l is the same** ($= \text{sup}(l)$); only confidence varies.
- Shortcut:** if $s \Rightarrow (l - s)$ fails, every rule whose antecedent is a *subset* of s also fails. Prune downward.

Worked example

PROBLEM not a PYQ · the rule half of 2081 Baishakh Q3 (b) verified/added

The frequent itemset $l = \{K, E, O\}$ has support count 3 in the 5-transaction database of Problem 3.1, where $\text{sup}(K) = 5$, $\text{sup}(E) = 4$, $\text{sup}(O) = 3$, $\text{sup}(K, E) = 4$, $\text{sup}(K, O) = 3$ and $\text{sup}(E, O) = 3$. List every association rule that l can generate and its confidence, and say which are strong at a minimum confidence of 80%.

From $l = \{K, E, O\}$ with $\text{sup} = 3$ (see Problem 3.1), $|l| = 3 \rightarrow 6$ candidate rules:

Rule	Confidence	Value	$\geq 80\%$?
$K \Rightarrow E, O$	3/5	0.60	×
$E \Rightarrow K, O$	3/4	0.75	×
$O \Rightarrow K, E$	3/3	1.00	✓
$K, E \Rightarrow O$	3/4	0.75	×
$K, O \Rightarrow E$	3/3	1.00	✓
$E, O \Rightarrow K$	3/3	1.00	✓

- Every rule with **O** in the antecedent succeeds, because O never appears without K and E.

M3. FP-tree construction and FP-Growth mining

Method

- Scan 1:** count items, drop infrequent ones, sort the rest by descending support $\rightarrow$ **F-list**. **State your tie-break rule.**
- Scan 2:** rewrite each transaction in F-list order, insert as a path, sharing prefixes and incrementing counts. Build the header table with node-links.
- Mine bottom-up** (least frequent item first): conditional pattern base $\rightarrow$ conditional FP-tree $\rightarrow$ recurse.

Problem 3.1 2+7

QUESTION AS PRINTED 2081 Baishakh · Q3 (b) · [2+7]

What is limitation of Apriori algorithm compared to FP-growth? A database has 5 transactions, given in table below. Let min support = 60% and min confidence = 80%.

TID	Item_bought
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I, E}

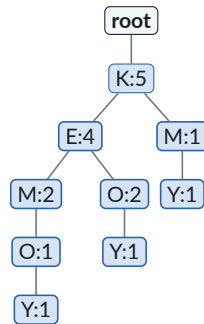
- Find all frequent itemsets using FP-growth.
 - List all of the strong association rules (with support and confidence).
- T500 prints O twice** — a transaction is a set, so it is {C, O, K, I, E}.
 - 5 transactions, 60% $\rightarrow$ **min count = 3**.

Step 1: counts and F-list

- K:5, E:4, M:3, O:3, Y:3 ✓ N:2, C:2, D:1, A:1, U:1, I:1 ×
- F-list = K(5), E(4), M(3), O(3), Y(3)** (ties at 3 broken alphabetically)

Step 2: reordered transactions

- T100 → K, E, M, O, Y
- T200 → K, E, O, Y
- T300 → K, E, M
- T400 → K, M, Y
- T500 → K, E, O

Step 3: the FP-tree

- **Check:** counts on each level sum correctly – K:5 = E:4 + M:1. ✓

Step 4: mine the tree bottom-up (FP-Growth proper — this is where the marks are)

- The procedure, for one item i at a time, taken in **reverse F-list order** (least frequent first, so $Y \rightarrow O \rightarrow M \rightarrow E \rightarrow K$):
 1. Follow i 's node-links to **every** node labelled i .
 2. For each such node, write down the **prefix path**: the labels from the root down to it, **excluding i itself**, and tag that whole path with **the count of the i -node**, not the counts written on the path. Together these are i 's **conditional pattern base**.
 3. Add up each label's tag totals across the base. Drop labels below min count. What remains is i 's **conditional FP-tree**.
 4. Every itemset in that conditional tree, with i appended, is frequent. Its support is the count in the conditional tree.
 5. If the conditional tree branches, recurse on it the same way.
- **The single most common mistake** is truncating a prefix path. A path is the *whole* route from the root, so an item sitting between the root and i stays in even when it is not adjacent to i .

Item	Conditional pattern base	Totals → cond. FP-tree	Frequent patterns
Y	{K,E,M,O}:1 {K,E,O}:1 {K,M}:1	K:3, E:2, M:2, O:2 → only K:3	{K,Y}:3
O	{K,E,M}:1 {K,E}:2	K:3, E:3, M:1 → K:3→E:3	{K,O}:3, {E,O}:3, {K,E,O}:3
M	{K,E}:2 {K}:1	K:3, E:2 → only K:3	{K,M}:3
E	{K}:4	K:4 → K:4	{K,E}:4
K	— (root child, no prefix)	—	{K}:5

- **Worked in full for Y**, since it is the one people get wrong. Y sits at the end of three different branches: $K \rightarrow E \rightarrow M \rightarrow O \rightarrow Y:1$, $K \rightarrow E \rightarrow O \rightarrow Y:1$ and $K \rightarrow M \rightarrow Y:1$. Strip the Y and keep **the whole remaining path each time**, so O belongs in the first two entries. Totals give K:3 only, hence the one pattern {K,Y}:3.
- **Why only O recurses:** its conditional tree still has two items (K and E), so the single branch $K:3 \rightarrow E:3$ yields {K,O}, {E,O} and the combination {K,E,O}. Every other conditional tree is a single node, so it yields exactly one pattern.

All frequent itemsets

- 1-itemsets: K:5, E:4, M:3, O:3, Y:3
- 2-itemsets: {K,E}:4, {K,M}:3, {K,O}:3, {E,O}:3, {K,Y}:3
- 3-itemsets: {K,E,O}:3

Strong rules (min conf 80%)**The node-link (header) table**

- One entry per F-list item, pointing at every node carrying that label:
 - K → K:5
 - E → E:4
 - M → M:2 (under E), M:1 (under K)
 - O → O:1 (under M), O:2 (under E)
 - Y → Y:1 × 3 (three different branches)
- **This table is what makes mining possible** — it is how you find all occurrences of an item without walking the tree.

- $K \Rightarrow E = 4/5 = 0.80 \checkmark$ $E \Rightarrow K = 4/4 = 1.0 \checkmark$
- $M \Rightarrow K = 3/3 = 1.0 \checkmark$ $O \Rightarrow K = 3/3 = 1.0 \checkmark$ $Y \Rightarrow K = 3/3 = 1.0 \checkmark$ $O \Rightarrow E = 3/3 = 1.0 \checkmark$
- $K, O \Rightarrow E = 1.0 \checkmark$ $E, O \Rightarrow K = 1.0 \checkmark$ $O \Rightarrow K, E = 1.0 \checkmark$
- Failing, all 7 of them: $K \Rightarrow M$, $K \Rightarrow O$, $K \Rightarrow Y$ and $K \Rightarrow E, O$ (all $3/5 = 0.6$), $E \Rightarrow O$, $K, E \Rightarrow O$ and $E \Rightarrow K, O$ (all $3/4 = 0.75$).
- **Count check:** five 2-itemsets give $5 \times 2 = 10$ rules and $\{K, E, O\}$ gives $2^3 - 2 = 6$, so **16 candidates = 9 strong + 7 failing.** ✓
- **Note K is on the right of five of the nine.** K is in every transaction, so anything implies it $\rightarrow$ high confidence, no information (§4.1).

Problem 3.2 4+4

QUESTION AS PRINTED 2075 Chaitra · Q4 · [4+4]

What do you mean by frequent Pattern growth, draw FP-tree with given tabular data.

TID	Items
01	f, a, c, d, g, i, m, p
02	a, b, c, f, l, m, o
03	b, f, h, j, o, w
04	b, c, k, s, p
05	A, f, c, e, l, p, m, n

Min support is not printed; use 3, which is the standard for this dataset. verified/added

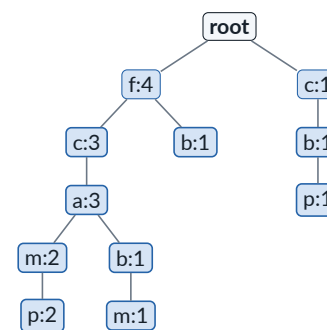
Counts and F-list

- f:4, c:4, a:3, b:3, m:3, p:3 ✓ l:2, o:2, and d, g, i, h, j, k, s, e, n, w all 1 ✗
- **F-list = f, c, a, b, m, p**
- **T05 prints a capital "A"** — same item as a.

Reordered transactions

- 01 $\rightarrow$ f, c, a, m, p 02 $\rightarrow$ f, c, a, b, m
- 03 $\rightarrow$ f, b 04 $\rightarrow$ c, b, p
- 05 $\rightarrow$ f, c, a, m, p

The FP-tree



- **Note the second root child:** transaction 04 starts with c, not f, so it cannot share the main prefix.

Problem 3.3 4+4

QUESTION AS PRINTED 2078 Bhadra · Q5 · [4+4]

Consider the given transactional database from a grocery store. Use a support threshold of 33.34% and confidence threshold of 60% to compute the following:

- Build a frequent pattern tree (FP-Tree). Show for each transaction how the tree evolves.
- Use FP-Growth algorithm to discover the frequent itemsets from this FP-tree.

Transaction ID	Items
T1	HotDogs, Buns, Ketchup
T2	HotDogs, Buns
T3	HotDogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	HotDogs, Coke, Chips

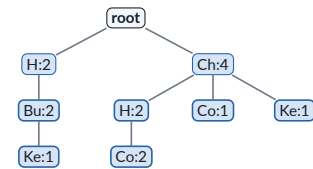
- 6 transactions, 33.34% $\rightarrow$ **min count = 2.**
- Counts: HotDogs 4, Chips 4, Coke 3, Buns 2, Ketchup 2 — **all frequent.**
- **F-list = Chips(4), HotDogs(4), Coke(3), Buns(2), Ketchup(2)** (4-way tie broken alphabetically).

Tree, built transaction by transaction

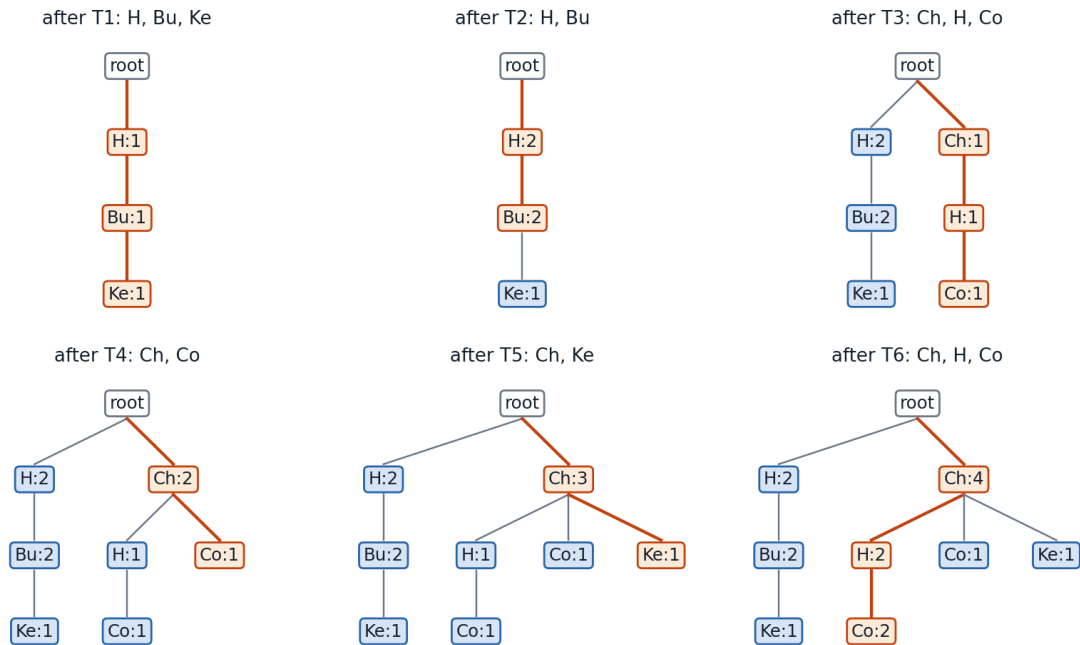
Final tree

T	Reordered	Effect on the tree
T1	H, Bu, Ke	new branch $H:1 \rightarrow Bu:1 \rightarrow Ke:1$
T2	H, Bu	prefix shared $\rightarrow H:2, Bu:2$
T3	Ch, H, Co	new root child $Ch:1 \rightarrow H:1 \rightarrow Co:1$
T4	Ch, Co	$Ch:2$, new child $Co:1$ under Ch
T5	Ch, Ke	$Ch:3$, new child $Ke:1$ under Ch
T6	Ch, H, Co	fully shared $\rightarrow Ch:4, H:2, Co:2$

- **Two root children**, because T1 and T2 do not contain Chips and so cannot join the Chips prefix.



“Show for each transaction how the tree evolves”: the six trees



The path each transaction touches is in orange: a new branch where no prefix is shared (T1, T3), incremented counts where it is (T2, T6). Drawn by `fpfig.py`, which rebuilds the tree from the transactions and checks every reordering and the final tree against this page.

Mining, least-frequent item first

Item	Conditional pattern base	Totals $\rightarrow$ patterns
Ketchup	{H, Bu}:1, {Ch}:1	H:1, Bu:1, Ch:1 $\rightarrow$ all below 2, no pattern
Buns	{H}:2	H:2 $\rightarrow$ {H, Bu}:2
Coke	{Ch, H}:2, {Ch}:1	Ch:3, H:2 $\rightarrow$ {Ch, Co}:3, {H, Co}:2, {Ch, H, Co}:2
HotDogs	{ }:2, {Ch}:2	Ch:2 $\rightarrow$ {Ch, H}:2
Chips	— (root child)	— $\rightarrow$ {Ch}:4

Frequent itemsets

- **1-itemsets**: Chips 4, HotDogs 4, Coke 3, Buns 2, Ketchup 2
- **2-itemsets**: {Chips, Coke} 3, {Chips, HotDogs} 2, {HotDogs, Coke} 2, {HotDogs, Buns} 2
- **3-itemsets**: {Chips, HotDogs, Coke} 2
- **HotDogs has an empty prefix once** (it is a root child from T1/T2), which contributes nothing to its conditional base. Write the empty set rather than omitting the row.

Problem 3.4 2+8

QUESTION AS PRINTED 2074 Chaitra · Q6 · [2+8]

How does FP growth approach overcomes the disadvantages of Apriori algorithm. For the transaction data given in table generate FP-Tree.

Transaction ID	Item set
T1	Camera, Laptop, Pen drive
T2	Laptop, Pen drive
T3	Laptop, Mobile, Earphone
T4	Earphone, Mobile
T5	Camera, Earphone
T6	Laptop, Mobile, Earphone

Min support is not printed; use 2. Abbreviations: E = Earphone, L = Laptop, Mo = Mobile, Ca = Camera, Pd = Pen drive.

Step 1: counts and F-list

Item	In	Count	Keep
Earphone	T3, T4, T5, T6	4	✓
Laptop	T1, T2, T3, T6	4	✓
Mobile	T3, T4, T6	3	✓
Camera	T1, T5	2	✓
Pen drive	T1, T2	2	✓

F-list = E(4), L(4), Mo(3), Ca(2), Pd(2) – **all five are frequent**, and the E/L tie at 4 is broken alphabetically.

Step 2: reorder each transaction by the F-list

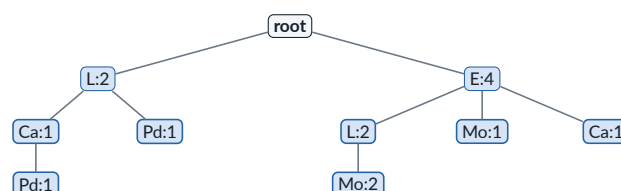
T	As printed	Reordered
T1	Camera, Laptop, Pen drive	L, Ca, Pd
T2	Laptop, Pen drive	L, Pd
T3	Laptop, Mobile, Earphone	E, L, Mo
T4	Earphone, Mobile	E, Mo
T5	Camera, Earphone	E, Ca
T6	Laptop, Mobile, Earphone	E, L, Mo

Step 3: insert them one at a time

T	Effect on the tree
T1	new branch L:1→Ca:1→Pd:1
T2	L:2, new child Pd:1 under L
T3	new root child E:1→L:1→Mo:1
T4	E:2, new child Mo:1 under E
T5	E:3, new child Ca:1 under E
T6	fully shared → E:4, L:2, Mo:2

- **Two root children:** T1 and T2 hold no Earphone, so they cannot join the E prefix.

Final FP-tree



- **Check:** L appears 4 times in D and the tree shows 2 under the root plus 2 under E. ✓ Do this for one item, it catches most construction errors.
- Mining it gives the key pattern {**Earphone, Laptop, Mobile**}:2 (T3 and T6). The question only asks for the tree, so this is a bonus line.

M4. Pattern evaluation: lift and correlation

Method

$$\text{lift}(A, B) = \frac{P(A \cup B)}{P(A)P(B)} = \frac{\text{conf}(A \Rightarrow B)}{\text{sup}(B)} \quad \chi^2 = \sum \frac{(\text{obs} - \text{exp})^2}{\text{exp}}$$

- $\text{lift} > 1$ positive · $= 1$ independent · < 1 negative.

Problem 4.1 8

QUESTION AS PRINTED 2070 Chaitra · Q5 · [8]

Why is pattern evaluation important in association rule mining? Explain with example the statistical based measures used for measuring interestingness of association rules.

The paper prints no data — the example is yours to supply. The standard *game* / *video* contingency table below is the one every textbook uses. verified/added

Given

10,000 transactions: 6,000 contain *game*, 7,500 contain *video*, 4,000 contain both.

Step 1: the support–confidence view says the rule is strong

- $\text{sup}(\text{game} \Rightarrow \text{video}) = 4000/10000 = 40\%$ (passes a 30% threshold)
- $\text{conf}(\text{game} \Rightarrow \text{video}) = 4000/6000 = 66.7\%$ (passes a 60% threshold)

Step 2: but look at the consequent alone

- $P(\text{video}) = 7500/10000 = 75\%$
- **Buying a game DROPS the chance of buying a video from 75% to 66.7%.** The rule is actively misleading.

Step 3: lift catches it

$$\text{lift} = \frac{0.40}{0.60 \times 0.75} = \frac{0.40}{0.45} = 0.89 < 1 \rightarrow \text{negatively correlated}$$

Step 4: χ^2 agrees

	game	<i>game</i>	row
video	4000 (4500)	3500 (3000)	7500
<i>video</i>	2000 (1500)	500 (1000)	2500
col	6000	4000	10,000

$$\chi^2 = \frac{500^2}{4500} + \frac{500^2}{3000} + \frac{500^2}{1500} + \frac{500^2}{1000} = 55.6 + 83.3 + 166.7 + 250 = 555.6$$

- Large $\chi^2 \rightarrow$ correlated; observed (4000) < expected (4500) $\rightarrow$ **negatively**.

Final answer

- **Pattern evaluation matters because support and confidence alone certify rules that are false.** Add a correlation measure before reporting any rule.
- Mention that lift and χ^2 are **not null-invariant**, so for sparse data prefer `all_conf`, `max_conf`, Kulczynski or cosine.

M5. Subsequence enumeration

Method

- A k -**subsequence** contains k items in total (not k elements).
- Choose how many items to take from each element, keeping element order; a chosen element must be non-empty to remain in the sequence.
- **Enumerate systematically** by the split (a, b, c, d) of items taken per element, then remove duplicates.

Problem 5.1 3

QUESTION AS PRINTED 2076 Chaitra · Q8 (a) · [3]

List all the 4-subsequences contained in the data sequence: $\langle \{1, 3\} \{2\} \{2, 3\} \{4\} \rangle$

Given

$e_1 = \{1, 3\}$, $e_2 = \{2\}$, $e_3 = \{2, 3\}$, $e_4 = \{4\}$; 6 items in total.

What counts as a subsequence (get this right first)

- Take a subset from each element, **possibly empty**. Elements chosen empty simply disappear.
- **Element order must be preserved**, and items from two different elements can never be merged into one element.
- A k -subsequence has k items in total, counted across all surviving elements. It is **not** k elements. This is the single most common misreading.
- So $\langle \{1, 3\} \{2\} \rangle$ is a 3-subsequence, not a 2-subsequence.

Why enumerate by splits

- A choice is fully described by **how many items are taken from each element**, written (a, b, c, d) for $e_1 \dots e_4$.
- The bounds are the element sizes: $a \leq 2$, $b \leq 1$, $c \leq 2$, $d \leq 1$, and we need $a + b + c + d = 4$.
- For a fixed split the number of ways is a product of binomials, $\binom{2}{a} \binom{1}{b} \binom{2}{c} \binom{1}{d}$, because the elements are chosen independently.
- **Enumerating splits first guarantees nothing is missed**, which listing by hand does not. Then remove duplicates at the end.

Step 1: enumerate the item-count splits

(a, b, c, d) with $a \leq 2$, $b \leq 1$, $c \leq 2$, $d \leq 1$, $\text{sum} = 4$

(a, b, c, d)	Ways	Subsequences
(2,1,1,0)	2	$\langle \{1, 3\} \{2\} \{2\} \rangle$, $\langle \{1, 3\} \{2\} \{3\} \rangle$
(2,1,0,1)	1	$\langle \{1, 3\} \{2\} \{4\} \rangle$
(2,0,2,0)	1	$\langle \{1, 3\} \{2, 3\} \rangle$
(2,0,1,1)	2	$\langle \{1, 3\} \{2\} \{4\} \rangle$ (duplicate), $\langle \{1, 3\} \{3\} \{4\} \rangle$
(1,1,2,0)	2	$\langle \{1\} \{2\} \{2, 3\} \rangle$, $\langle \{3\} \{2\} \{2, 3\} \rangle$
(1,1,1,1)	4	$\langle \{1\} \{2\} \{2\} \{4\} \rangle$, $\langle \{1\} \{2\} \{3\} \{4\} \rangle$, $\langle \{3\} \{2\} \{2\} \{4\} \rangle$, $\langle \{3\} \{2\} \{3\} \{4\} \rangle$
(1,0,2,1)	2	$\langle \{1\} \{2, 3\} \{4\} \rangle$, $\langle \{3\} \{2, 3\} \{4\} \rangle$
(0,1,2,1)	1	$\langle \{2\} \{2, 3\} \{4\} \rangle$

Step 2: check the count, then remove duplicates

- Ways per split: $2 + 1 + 1 + 2 + 2 + 4 + 2 + 1 = 15$ selections. **Do this arithmetic** — it is how you know no split was dropped.
- **A duplicate appears when the same items can be taken from different elements**. Here $e_2 = \{2\}$ and $e_3 = \{2, 3\}$ share the item 2, so $\langle \{1, 3\} \{2\} \{4\} \rangle$ is produced twice: once as (2, 1, 0, 1) taking the 2 from e_2 , once as (2, 0, 1, 1) taking it from e_3 . **Same sequence, so it counts once**.
- No other pair of splits can collide, because every other split keeps a different number of surviving elements or a different element content.

Final answer

- 15 selections — 1 duplicate → **14 distinct 4-subsequences**.
- **The exam usually only wants “at least four”** 79 Bh, so listing the first few from a clearly stated method is enough. **Show the method, not just the list.**

Problem 5.2 3

QUESTION AS PRINTED 2079 Bhadra · Q6 (a) · [3+3]

How do you handle the categorical attributes in data mining process? Explain with example. Generate the at least four subsequences from the given sequence: $\langle \{2, 3, 5\}, \{6, 7, 8\}, \{9, 1\}, \{7, 4\} \rangle$.

- Any order-preserving selection works. Four valid examples:
 - $\langle \{2\} \{6\} \rangle$ $\langle \{3, 5\} \{7\} \{9\} \rangle$
 - $\langle \{2, 3\} \{6, 7\} \{1\} \{4\} \rangle$ $\langle \{5\} \{8\} \{9, 1\} \{7\} \rangle$
- **Must hold:** element order preserved, and each chosen sub-element is a subset of the original element. **Must not do:** reorder elements, or merge items from two elements into one.

M6. Subgraph candidate generation by edge growing

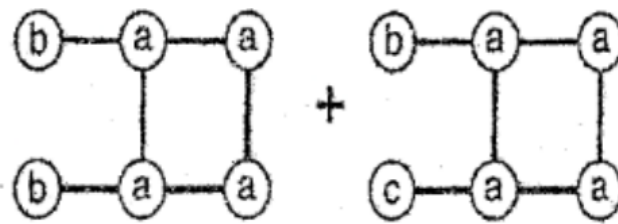
Method

- Two frequent subgraphs are joined only if they share a **common core**: the subgraph left when one edge is removed from each.
- Edge growing** joins two k -edge graphs sharing a $(k - 1)$ -edge core and produces a $(k + 1)$ -**edge candidate** = core + the edge unique to each parent.
- Vertex growing** instead adds a vertex, and generates far more candidates. **Edge growing is preferred for exactly that reason.**
- The trap, and the whole point of this question:** when the core carries **repeated vertex labels**, the two extra edges can be attached in more than one non-isomorphic way, so **one join yields several candidates**. Enumerate them, do not assume there is one.

Problem 6.1 3

QUESTION AS PRINTED 2076 Chaitra · Q8 (b) · [3]

Draw all candidate sub-graphs obtained from joining the pair of graphs shown below using edge-growing method to expand the sub-graphs.



Step 1: read the two parents

- Both are a **4-cycle of a vertices** with two pendant vertices hanging off two *adjacent* corners of that cycle.
- G_1 : pendants are b and b . G_2 : pendants are b and c .
- Each has **6 vertices and 6 edges**, so $k = 6$.

Step 2: find the core

- Delete one edge from each so that what is left is the same graph: drop the second pendant edge from both.
- Core = 4-cycle of a 's + one pendant b** , 5 vertices and 5 edges = $k - 1$. ✓ the graphs are joinable.
- Edge unique to G_1 : $a - b$. Edge unique to G_2 : $a - c$.

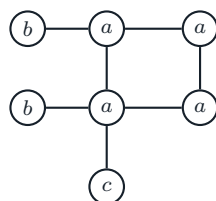
Step 3: the candidate size

- Candidate = core + both unique edges $\rightarrow 5 + 2 = 7$ edges, 7 vertices.

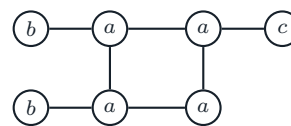
Step 4: where can the two new edges attach?

- Call the core's pendant-carrying a vertex A_1 . In *both* parents the second pendant sits on a cycle vertex **adjacent to A_1** , and A_1 has two neighbours in the cycle, A_2 and A_4 .
- All four placements (b, c) are legal: (A_2, A_2) , (A_4, A_4) , (A_2, A_4) , (A_4, A_2) .
- The core has a **mirror symmetry swapping A_2 and A_4** , so these collapse to **two non-isomorphic candidates**:
 $(A_2, A_2) \cong (A_4, A_4) \rightarrow$ **candidate I**
 $(A_2, A_4) \cong (A_4, A_2) \rightarrow$ **candidate II**
- This multiplicity comes entirely from the four a 's sharing one label.** Had the cycle been a, b, c, d the join would have given a single candidate.

Final answer: 2 candidate sub-graphs, each with 7 vertices and 7 edges



Candidate I
both new pendants
on the same a



Candidate II
new pendants on the two
different neighbours of A_1

- Marking note:** the marks are for showing the *core* and then arguing the placements. A single drawing with no core and no count of candidates scores badly even if the drawing itself is one of the two right answers.
- Cross-check:** every candidate must contain both parents as subgraphs, and must have exactly $k + 1 = 7$ edges. Both do.

Ch 5 – Cluster Analysis – Numerical Problems & Solutions

3 methods · 11 PYQ instances · every distance recomputed. K-means appears in 7 papers and hierarchical dendrograms in 4, so these two methods alone are worth around 60 marks across the archive.

Methods

- **M1** K-means 78 Bh, 74 Ash, 75 Ash, 71 Shr, 76 Ch, 81 Ba
- **M2** Agglomerative hierarchical clustering + dendrogram 80 Bh, 80 Ba, 76 Ch
- **M3** DBSCAN point classification *practice*

M1. K-means

Method

1. State K and the initial centroids. **Always write down which points you seeded with** — the answer depends on it and the examiner needs to follow you.
 2. Build a distance table: every point against every centroid.
 3. Assign each point to its nearest centroid.
 4. Recompute each centroid as the **mean** of its members.
 5. Repeat until the assignment does not change. **That repetition is the convergence proof.**
- You may compare *squared* distances to avoid square roots — the ordering is identical. Say that you are doing it.

Problems 1.1 and 1.2 2+6 2+8

QUESTION AS PRINTED 2078 Bhadra · Q8 · [2+6]

Write K-means clustering algorithm. Generate two clusters from following dataset using K-means clustering.

Instance	A	B
1	1	2
2	2.5	1
3	3.5	1.5
4	4	1
5	3.5	2.5
6	5	3

QUESTION AS PRINTED 2074 Ashwin · Q7 · [2+8]

Write K-means algorithm and find clusters for following data set.

Instance	X	Y
1	1.0	2.0
2	2.5	1.0
3	3.5	1.5
4	4.0	1.0
5	3.5	2.5
6	5.0	3.0

(Take $K = 2$)

The same 6 instances, only the column names differ.

Seeds

Take the two **most widely separated** points, $c_1 = p_1(1, 2)$ and $c_2 = p_6(5, 3)$. (A standard, defensible heuristic; state it.)

Iteration 1

Point	d to $c_1(1, 2)$	d to $c_2(5, 3)$	Cluster
$p_1(1, 2)$	0	4.123	C1
$p_2(2.5, 1)$	1.803	3.202	C1
$p_3(3.5, 1.5)$	2.550	2.121	C2
$p_4(4, 1)$	3.162	2.236	C2
$p_5(3.5, 2.5)$	2.550	1.581	C2
$p_6(5, 3)$	4.123	0	C2

- $C1 = \{p_1, p_2\} \rightarrow$ new centroid $(\frac{1+2.5}{2}, \frac{2+1}{2}) = (1.75, 1.5)$
- $C2 = \{p_3, p_4, p_5, p_6\} \rightarrow$ new centroid $(\frac{3.5+4+3.5+5}{4}, \frac{1.5+1+2.5+3}{4}) = (4, 2)$

Iteration 2

Point	d to $(1.75, 1.5)$	d to $(4, 2)$	Cluster
p_1	0.901	3.000	C1
p_2	0.901	1.803	C1
p_3	1.750	0.707	C2
p_4	2.305	1.000	C2
p_5	2.016	0.707	C2
p_6	3.580	1.414	C2

- **Assignment unchanged $\rightarrow$ converged.**

Final answer

- **Cluster 1** = $\{(1, 2), (2.5, 1)\}$, centroid $(1.75, 1.5)$
- **Cluster 2** = $\{(3.5, 1.5), (4, 1), (3.5, 2.5), (5, 3)\}$, centroid $(4, 2)$
- Converged in **2 iterations**.

Seed sensitivity – worth a line, since the question asks for limitations. Seeding instead with the *first* two points, $c_1 = p_1(1, 2)$ and $c_2 = p_2(2.5, 1)$, gives $C1 = \{p_1\}$ and $C2 = \{p_2, p_3, p_4, p_5, p_6\}$, which also converges but leaves a **degenerate 1-point cluster**.

Same data, same K , **different answer** – this is exactly the “sensitive to initial centroids / converges to a local optimum” limitation. verified/added

Problem 1.3 4+6

QUESTION AS PRINTED 2075 Ashwin · Q7 · [4+6]

Explain K-means clustering with limitation. Generate two clusters from following dataset using K-means clustering.

A	B
1	2
2.5	4.5
4	6
3.5	4
4	5.5
3	6

Seeds

$c_1 = (1, 2)$ (clearly isolated) and $c_2 = (4, 6)$. (State the choice.)

Iteration 1

Point	d to $c_1(1, 2)$	d to $c_2(4, 6)$	Cl.
$(1, 2)$	0	5.000	C1
$(2.5, 4.5)$	2.915	2.121	C2
$(4, 6)$	5.000	0	C2
$(3.5, 4)$	3.202	2.062	C2
$(4, 5.5)$	4.610	0.500	C2
$(3, 6)$	4.472	1.000	C2

- $C1 = \{(1, 2)\} \rightarrow$ centroid stays **(1, 2)**
- $C2 \rightarrow$ centroid $(\frac{2.5+4+3.5+4+3}{5}, \frac{4.5+6+4+5.5+6}{5}) = (3.4, 5.2)$

Iteration 2

Point	d to $(1, 2)$	d to $(3.4, 5.2)$	Cl.
$(1, 2)$	0	4.000	C1
$(2.5, 4.5)$	2.915	1.140	C2
$(4, 6)$	5.000	1.000	C2
$(3.5, 4)$	3.202	1.204	C2
$(4, 5.5)$	4.610	0.671	C2
$(3, 6)$	4.472	0.894	C2

- **Assignment unchanged $\rightarrow$ converged** after 2 iterations.

Final answer

$C1 = \{(1, 2)\}$, $C2 = \{(2.5, 4.5), (4, 6), (3.5, 4), (4, 5.5), (3, 6)\}$, centroid $(3.4, 5.2)$.

- **Note:** $(1, 2)$ is a genuine outlier here, so the 1-point cluster is the honest answer, not a seeding artefact. Say so, and add that **K-medoids** or **DBSCAN** would treat it as noise.

Problem 1.4 10

QUESTION AS PRINTED 2071 Shrawan · Q8 · [10]

Explain K-means clustering with limitation Use k-means clustering to cluster the following dataset.

A	B
1.0	1.0
1.5	2.0
3.0	4.0
5.0	7.0
3.5	5.0
4.5	5.0
3.5	4.5

The paper prints no K ; take $K = 2$, the value its sister questions use.

Seeds

$c_1 = (1, 1)$, $c_2 = (5, 7)$ – the two most widely separated points.

Iteration 1

Point	d to $c_1(1, 1)$	d to $c_2(5, 7)$	Cl.
(1, 1)	0	7.211	C1
(1.5, 2)	1.118	6.104	C1
(3, 4)	3.606	3.606	tie
(5, 7)	7.211	0	C2
(3.5, 5)	4.717	2.500	C2
(4.5, 5)	5.315	2.062	C2
(3.5, 4.5)	4.301	2.915	C2

- **(3, 4) is an exact tie:** $\sqrt{4+9} = \sqrt{4+9} = 3.606$ both ways. **State a tie-break rule and use it** – here, send it to C2. Examiners give the mark for naming the rule, not for the direction chosen.
- C1 → centroid **(1.25, 1.5)**
- C2 → $\left(\frac{3+5+3.5+4.5+3.5}{5}, \frac{4+7+5+5+4.5}{5}\right) = \mathbf{(3.9, 5.1)}$

Final answer

$C1 = \{(1, 1), (1.5, 2)\}$ centroid $(1.25, 1.5)$; $C2 = \{(3, 4), (5, 7), (3.5, 5), (4.5, 5), (3.5, 4.5)\}$ centroid $(3.9, 5.1)$.

Problem 1.5 8

QUESTION AS PRINTED 2076 Chaitra · Q9 · [8]

Given the matrix X whose rows represent different data points, perform a k-means clustering on this dataset using the Euclidean distance as the distance function. Here K is chosen as 3. The center of the 3 clusters are initialized as red $(6.2, 3.2)$, green $(6.6, 3.7)$ and blue $(6.5, 3.0)$. Provide the final cluster centers and comment on the number of iterations required for the clusters to converge.

$$X = \begin{bmatrix} 5.9 & 3.2 \\ 4.6 & 2.9 \\ 6.2 & 2.8 \\ 4.7 & 3.2 \\ 5.5 & 4.2 \\ 5.0 & 3.0 \\ 4.9 & 3.1 \\ 6.7 & 3.1 \\ 5.1 & 3.8 \\ 6.0 & 3.0 \end{bmatrix}$$

Iteration 1

(assign to nearest of the 3 given centres)

- **red** $\{p_1, p_2, p_4, p_6, p_7, p_9, p_{10}\}$ **green** $\{p_5\}$ **blue** $\{p_3, p_8\}$
- New centres: red (5.171, 3.171), green (5.5, 4.2), blue (6.45, 2.95)

Iteration 2

- **red** $\{p_2, p_4, p_6, p_7\}$ **green** $\{p_5, p_9\}$ **blue** $\{p_1, p_3, p_8, p_{10}\}$
- New centres: red (4.8, 3.05), green (5.3, 4.0), blue (6.2, 3.025)

Iteration 3

- Every point stays where it was → **converged**.

Final answer

Cluster	Members	Final centre
red	p_2, p_4, p_6, p_7	(4.80, 3.05)
green	p_5, p_9	(5.30, 4.00)
blue	p_1, p_3, p_8, p_{10}	(6.20, 3.025)

- **Comment on iterations:** assignments changed at iterations 1 and 2 and were stable at iteration 3, so **convergence took 3 iterations (2 real updates + 1 confirming pass)**.
- Worth adding: the 3 supplied centres were all bunched on the right of the data ($x \approx 6.2$ – 6.6), yet K-means still pulled them apart into a sensible left / middle / right split. Good illustration that it converges quickly even from a poor start.

Problem 1.6 – the defective question 5+3

QUESTION AS PRINTED 2081 Baishakh · Q4 (b) · [5+3]

Use K-means clustering to cluster the following given data for $K = 2$ with Euclidean distance matrix. List down the demerits of this algorithm.

(nothing follows on the paper – the next line is Q5)

The paper prints NO data table for this question – it jumps straight from the sentence to Q5. Verified against the original scan and recorded in the PYQ archive.

In the exam: answer the 3-mark demerits part in full (\$5.2), write the 5 steps of the algorithm, and demonstrate on any small dataset of your own, stating that you assumed one. Use **Problem 1.1's** 6 points – it is the dataset this paper's sister questions use. **verified/added**

Problem 1.7 2+8

QUESTION AS PRINTED 2082 Bhadra · Q6 · [2+8]

Describe the K-means clustering algorithm. Generate two clusters from following dataset using K-means clustering.

Instance	A	B
1	1	2
2	1.5	1
3	3.5	1.5
4	4	3
5	3.5	2.5
6	6	4

Seeds

the two most widely separated points, $c_1 = (1, 2)$ and $c_2 = (6, 4)$.

Iteration 1

Point	d to (1, 2)	d to (6, 4)	Cl.
(1, 2)	0	5.385	C1
(1.5, 1)	1.118	5.408	C1
(3.5, 1.5)	2.550	3.536	C1
(4, 3)	3.162	2.236	C2
(3.5, 2.5)	2.550	2.915	C1
(6, 4)	5.385	0	C2
- C1 $\rightarrow \left(\frac{1+1.5+3.5+3.5}{4}, \frac{2+1+1.5+2.5}{4} \right) = (2.375, 1.75)$			
- C2 $\rightarrow \left(\frac{4+6}{2}, \frac{3+4}{2} \right) = (5, 3.5)$			

Final answer

- **Cluster 1** = {(1, 2), (1.5, 1), (3.5, 1.5), (3.5, 2.5)}, centroid (2.375, 1.75)
- **Cluster 2** = {(4, 3), (6, 4)}, centroid (5, 3.5)

Problem 1.8 5+3

QUESTION AS PRINTED 2081 Bhadra · Q8 · [5+3]

Consider the data points {(2, 3), (3, 3), (6, 8), (8, 8), (7, 5)} with initial centroid (2, 3). Apply K-means algorithm to choose the next centroid and cluster these data points into $k = 2$. Discuss the problems that arise with K-means clustering algorithm and ways to solve it.

Only ONE centroid is given, and “choose the next centroid” is the first thing asked. Take the farthest point from the given one — the standard max-separation seeding rule — and say so.

Step 1: choose the second centroid

Distance from (2, 3)	(3, 3)	(6, 8)	(8, 8)	(7, 5)
Euclidean	1.000	6.403	7.810	5.385

(8, 8) is farthest $\rightarrow c_2 = (8, 8)$.

Iteration 1

Point	d to (2, 3)	d to (8, 8)	Cl.
(2, 3)	0	7.810	C1
(3, 3)	1.000	7.071	C1
(6, 8)	6.403	2.000	C2
(8, 8)	7.810	0	C2
(7, 5)	5.385	3.162	C2
- C1 $\rightarrow (2.5, 3)$ C2 $\rightarrow \left(\frac{6+8+7}{3}, \frac{8+8+5}{3} \right) = (7, 7)$			

Final answer

C1 = {(2, 3), (3, 3)} centroid (2.5, 3); C2 = {(6, 8), (8, 8), (7, 5)} centroid (7, 7). The 3-mark “problems and ways to solve” half is the limitations list in §5.2.

Problem 1.9 4+8

QUESTION AS PRINTED 2073 Chaitra · Q7 · [4+8]

How clustering differ from classification? Given the one-dimensional points {5, 12, 18, 24, 30, 42, 48} with initial centroids {5, 12, 18}, create three clusters by K-Means algorithm and calculate SSE for this clustering result.

One dimension, so “distance” is just $|x - c|$. This is the only PYQ that asks for SSE, and it is the only one that runs past two iterations.

Iter.	Assignment (nearest centroid)	Clusters	New centroids
1	centroids 5, 12, 18	{5}, {12}, {18, 24, 30, 42, 48}	5, 12, $\frac{162}{5} = 32.4$
2	centroids 5, 12, 32.4	{5}, {12, 18}, {24, 30, 42, 48}	5, 15, $\frac{144}{4} = 36$
3	centroids 5, 15, 36	{5}, {12, 18, 24}, {30, 42, 48}	5, 18, 40
4	centroids 5, 18, 40	{5}, {12, 18, 24}, {30, 42, 48}	5, 18, 40 unchanged

- **Converged at iteration 4**. Note the cluster boundaries kept moving for three rounds — do not stop after two just because the earlier problems did.

SSE

$$SSE = \sum_i \sum_{x \in C_i} (x - c_i)^2$$

Cluster	Centroid	Squared errors	Subtotal
{5}	5	$(5 - 5)^2 = 0$	0
{12, 18, 24}	18	$(12 - 18)^2 + (18 - 18)^2 + (24 - 18)^2 = 36 + 0 + 36$	72
{30, 42, 48}	40	$(30 - 40)^2 + (42 - 40)^2 + (48 - 40)^2 = 100 + 4 + 64$	168

$$SSE = 0 + 72 + 168 = 240$$

- **SSE is the objective K-means minimises**, so quoting it is how you show the clustering is the algorithm's answer and not just a guess. A different seeding that gave a larger SSE would be a worse local optimum.

Problem 1.10 – the Manhattan one 8

QUESTION AS PRINTED 2079 Baishakh · Q8 · [8]

Define clustering. Given the matrix whose column represent different data points, perform a K-means clustering on this dataset using the **Manhattan** as the distance function. The center of the 3 clusters are initiated as $A(6.2, 3.2)$, $B(6.6, 3.7)$ and $C(6.5, 3.0)$. Provide the final cluster centers after 3 iteration.

x	5.9	4.6	6.2	4.7	5.5	5.0	4.9	6.7	5.1	6.0
y	3.2	2.9	2.8	3.2	4.2	3.0	3.1	3.1	3.8	3.0

Manhattan, not Euclidean: $d = |x_1 - x_2| + |y_1 - y_2|$, no squares and no square root. **Stop after 3 iterations** – the question says so, and it has *not* converged by then. Say that in the answer.

Iteration 1centres $A(6.2, 3.2)$, $B(6.6, 3.7)$, $C(6.5, 3.0)$

Point	d to A	d to B	d to C	Cluster
(5.9, 3.2)	0.30	1.20	0.80	A
(4.6, 2.9)	1.90	2.80	2.00	A
(6.2, 2.8)	0.40	1.30	0.50	A
(4.7, 3.2)	1.50	2.40	2.00	A
(5.5, 4.2)	1.70	1.60	2.20	B
(5.0, 3.0)	1.40	2.30	1.50	A
(4.9, 3.1)	1.40	2.30	1.70	A
(6.7, 3.1)	0.60	0.70	0.30	C
(5.1, 3.8)	1.70	1.60	2.20	B
(6.0, 3.0)	0.40	1.30	0.50	A

New centres: $A = (5.329, 3.029)$ (7 points), $B = (5.3, 4.0)$ (2), $C = (6.7, 3.1)$ (1).

Iteration 2centres $A(5.329, 3.029)$, $B(5.3, 4.0)$, $C(6.7, 3.1)$

Point	d to A	d to B	d to C	Cluster
(5.9, 3.2)	0.74	1.40	0.90	A
(4.6, 2.9)	0.86	1.80	2.30	A
(6.2, 2.8)	1.10	2.10	0.80	C
(4.7, 3.2)	0.80	1.40	2.10	A
(5.5, 4.2)	1.34	0.40	2.30	B
(5.0, 3.0)	0.36	1.30	1.80	A
(4.9, 3.1)	0.50	1.30	1.80	A
(6.7, 3.1)	1.44	2.30	0.00	C
(5.1, 3.8)	1.00	0.40	2.30	B
(6.0, 3.0)	0.70	1.70	0.80	A

New centres: $A = (5.183, 3.067)$ (6), $B = (5.3, 4.0)$ (2), $C = (6.45, 2.95)$ (2).

Iteration 3centres $A(5.183, 3.067)$, $B(5.3, 4.0)$, $C(6.45, 2.95)$

Point	d to A	d to B	d to C	Cluster
(5.9, 3.2)	0.85	1.40	0.80	C
(4.6, 2.9)	0.75	1.80	1.90	A
(6.2, 2.8)	1.28	2.10	0.40	C
(4.7, 3.2)	0.62	1.40	2.00	A
(5.5, 4.2)	1.45	0.40	2.20	B
(5.0, 3.0)	0.25	1.30	1.50	A
(4.9, 3.1)	0.32	1.30	1.70	A
(6.7, 3.1)	1.55	2.30	0.40	C
(5.1, 3.8)	0.82	0.40	2.20	B
(6.0, 3.0)	0.88	1.70	0.50	C

Final answer after 3 iterations

- $A = (4.8, 3.05)$ from $\{(4.6, 2.9), (4.7, 3.2), (5.0, 3.0), (4.9, 3.1)\}$
- $B = (5.3, 4.0)$ from $\{(5.5, 4.2), (5.1, 3.8)\}$
- $C = (6.2, 3.025)$ from $\{(5.9, 3.2), (6.2, 2.8), (6.7, 3.1), (6.0, 3.0)\}$
- **The assignment was still changing at iteration 3**, so this is *not* a converged clustering. The question asked for 3 iterations, so stop there and **state that it has not converged** – that sentence is worth a mark.
- **Note how the three seeds were all bunched near (6.5, 3)**, which is why cluster C started with a single point and had to grow every round. A textbook illustration of **poor seeding**.

M2. Agglomerative hierarchical clustering

Method

1. Start with each point as its own cluster; write the distance matrix.
 2. Find the **smallest** entry $\rightarrow$ merge those 2 clusters. **Record the merge height** – that value is the dendrogram bar.
 3. Update the matrix using the linkage rule: **single** = MIN of the two rows, **complete** = MAX.
 4. Repeat until one cluster remains, then draw the dendrogram with merges at their heights.
- **Merge heights never decrease.** If yours does, you have made an arithmetic slip.
 - n points always give exactly $n - 1$ merges.

Problem 2.1 2+7

QUESTION AS PRINTED 2080 Bhadra · Q4 (b) · [2+7]

What is hierarchical clustering? Use this clustering approach to draw dendrogram for given data points.

	p_1	p_2	p_3	p_4	p_5	p_6
p_1	0.00	0.24	0.22	0.37	0.34	0.23
p_2	0.24	0.00	0.15	0.20	0.14	0.25
p_3	0.22	0.15	0.00	0.15	0.28	0.11
p_4	0.37	0.20	0.15	0.00	0.29	0.22
p_5	0.34	0.14	0.28	0.29	0.00	0.39
p_6	0.23	0.25	0.11	0.22	0.39	0.00

Distances are given directly, so no coordinate work. The paper does not name a linkage – use **single link (MIN)** and say so. **Recompute and rewrite the whole matrix after every merge** – that redrawing is the method, and it is where the marks are.

Step 1

smallest entry in the matrix above is $d(p_3, p_6) = 0.11 \rightarrow$ merge at height **0.11**

Matrix after merge 1

5 clusters

	p_1	p_2	p_4	p_5
p_1	0.00	0.24	0.37	0.34
p_2	0.24	0.00	0.20	0.14
p_4	0.37	0.20	0.00	0.29
p_5	0.34	0.14	0.29	0.00
$\{p_3p_6\}$	0.22	0.15	0.15	0.28

Step 2smallest is now $d(p_2, p_5) = 0.14 \rightarrow$ merge at height **0.14****Matrix after merge 2**

4 clusters

	p_1	p_4	$\{p_3p_6\}$
p_1	0.00	0.37	0.22
p_4	0.37	0.00	0.15
$\{p_3p_6\}$	0.22	0.15	0.00
$\{p_2p_5\}$	0.24	0.20	0.15

Step 3take $d(\{p_3p_6\}, p_4) = 0.15 \rightarrow$ merge at height **0.15****Matrix after merge 3**

3 clusters

	p_1	$\{p_2p_5\}$
p_1	0.00	0.24
$\{p_2p_5\}$	0.24	0.00
$\{p_3p_4p_6\}$	0.22	0.15

Step 4smallest is **0.15** $\rightarrow$ merge $\{p_3p_4p_6\}$ with $\{p_2p_5\}$ at height **0.15****Matrix after merge 4**

2 clusters

	p_1
p_1	0.00
$\{p_2p_3p_4p_5p_6\}$	0.22

Step 5only one distance left $\rightarrow$ merge everything at height **0.22****Final answer**Dendrogram heights: **0.11, 0.14, 0.15, 0.15, 0.22**.

- Reading it: $\{p_3, p_6\}$ is the tightest pair; p_1 is the most isolated point, joining last.
- Cutting between 0.15 and 0.22 gives **2 clusters**: $\{p_1\}$ and $\{p_2, p_3, p_4, p_5, p_6\}$.

How the new row was computed

MIN of the two old rows

- $d(\{p_3p_6\}, p_1) = \min(0.22, 0.23) = 0.22$
- $d(\{p_3p_6\}, p_2) = \min(0.15, 0.25) = 0.15$
- $d(\{p_3p_6\}, p_4) = \min(0.15, 0.22) = 0.15$
- $d(\{p_3p_6\}, p_5) = \min(0.28, 0.39) = 0.28$
- Every other entry is copied over unchanged.

New row

- $d(\{p_2p_5\}, p_1) = \min(0.24, 0.34) = 0.24$
- $d(\{p_2p_5\}, p_4) = \min(0.20, 0.29) = 0.20$
- $d(\{p_2p_5\}, \{p_3p_6\}) = \min(0.15, 0.28) = 0.15$
- **A tie at 0.15**, between $(\{p_3p_6\}, p_4)$ and $(\{p_3p_6\}, \{p_2p_5\})$.
Either may be merged first and the dendrogram heights come out the same; state the rule you use. Taken here in the order the matrix is read.

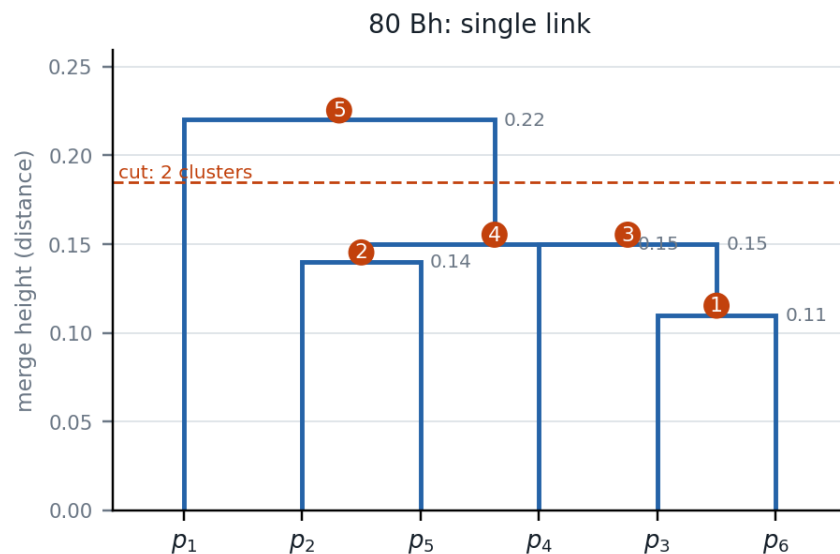
New row

- $d(\{p_3p_4p_6\}, p_1) = \min(0.22, 0.37) = 0.22$
- $d(\{p_3p_4p_6\}, \{p_2p_5\}) = \min(0.15, 0.20) = 0.15$

New row

- $d(\{p_2p_3p_4p_5p_6\}, p_1) = \min(0.24, 0.22) = 0.22$, the smaller of the two surviving links to p_1 .

The dendrogram



Bars at the five merge heights, numbered in merge order (Steps 1 to 5 above); merges 3 and 4 share the height 0.15, the tie. The dashed line is the 2-cluster cut. Drawn by `dendfig.py`, which replays the merges and checks them against this page.

Problem 2.2 8

QUESTION AS PRINTED 2080 Baishakh · Q7 · [8]

Cluster the following samples based on complete-linkage algorithm and draw the dendrogram. Using Euclidean distance.

Point	x Coordinate	y Coordinate
p1	0.40	0.53
p2	0.22	0.38
p3	0.35	0.32
p4	0.26	0.19
p5	0.08	0.41
p6	0.45	0.30

Step 1: Euclidean distance matrix

$p_1(0.40, 0.53)$ $p_2(0.22, 0.38)$ $p_3(0.35, 0.32)$ $p_4(0.26, 0.19)$ $p_5(0.08, 0.41)$ $p_6(0.45, 0.30)$

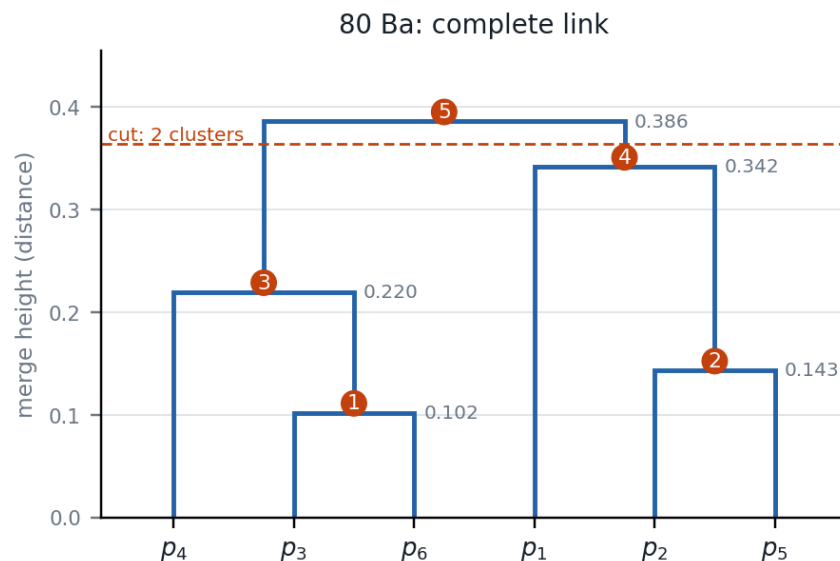
	p_1	p_2	p_3	p_4	p_5
p_2	0.234				
p_3	0.216	0.143			
p_4	0.368	0.194	0.158		
p_5	0.342	0.143	0.285	0.284	
p_6	0.235	0.244	0.102	0.219	0.386

Step 2: merge sequence (MAX update)

Step	Merge	Height	Updated distances (MAX)
1	p_3, p_6	0.102	to p_1 0.235, p_2 0.244, p_4 0.219, p_5 0.386
2	p_2, p_5	0.143	to p_1 0.342, p_4 0.284, $\{p_3p_6\}$ 0.386
3	$\{p_3p_6\}, p_4$	0.219	to p_1 0.368, $\{p_2p_5\}$ 0.386
4	$p_1, \{p_2p_5\}$	0.342	to $\{p_3p_4p_6\}$ 0.386
5	the two remaining	0.386	—

Final answer

Heights **0.102, 0.143, 0.219, 0.342, 0.386**; cutting below 0.386 gives $\{p_1, p_2, p_5\}$ and $\{p_3, p_4, p_6\}$.



The dendrogram the question asks for, merges numbered as in the Step 2 table; the cut below 0.386 leaves the two groups. From `dendfig.py`.

Problem 2.3 4+4

QUESTION AS PRINTED 2076 Chaitra · Q10 · [4+4]

The table below is a distance matrix for six objects:

	A	B	C	D	E	F
A	0					
B	0.12	0				
C	0.51	0.25	0			
D	0.84	0.16	0.14	0		
E	0.28	0.77	0.70	0.45	0	
F	0.34	0.61	0.93	0.20	0.67	0

- Show the final result of hierarchical clustering with single-link by drawing a dendrogram.
- Show the final result of hierarchical clustering with complete-link by drawing a dendrogram.

The ideal question for showing you understand the difference — one matrix, both linkages.

(a) Single link (MIN)

	Merge	Height
1	A, B	0.12
2	C, D	0.14
3	{AB}, {CD}	0.16
4	{ABCD}, F	0.20
5	all, + E	0.28

- Heights: **0.12, 0.14, 0.16, 0.20, 0.28**
- Everything merges early and at similar heights → chaining.

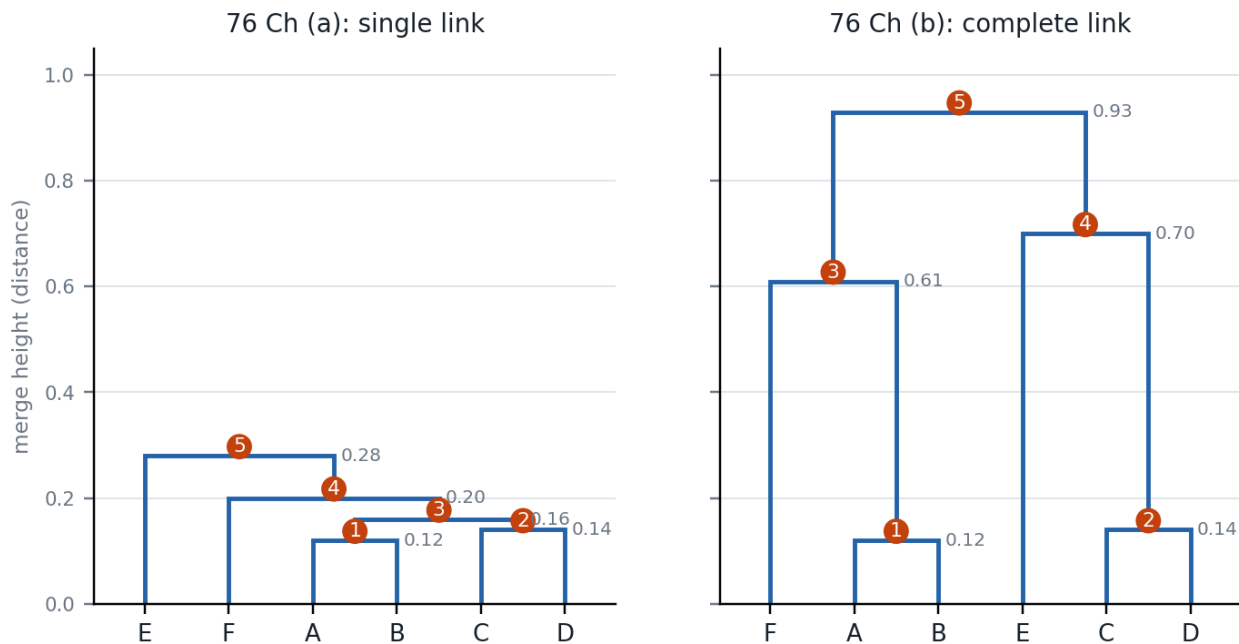
(b) Complete link (MAX)

	Merge	Height
1	A, B	0.12
2	C, D	0.14
3	{AB}, F	0.61
4	{CD}, E	0.70
5	the two groups	0.93

- Heights: **0.12, 0.14, 0.61, 0.70, 0.93**
- Final 2 clusters: **{A,B,F}** and **{C,D,E}** — a balanced split.

The comparison is the marks. The first 2 merges are identical, then the methods diverge completely: single link strings everything onto the A-B-C-D chain and leaves E as a straggler, while complete link produces two clean balanced groups. **Single link** chases the nearest neighbour, so it follows chains and elongated shapes. **Complete link** is held back by the worst pair, so it prefers compact globular clusters. verified/added

Both dendrograms, to one scale



(a) single link: everything joins below 0.3, the chaining in one look. (b) complete link: two compact groups that only meet at 0.93. Merges numbered as in the two tables. From `dendfig.py`.

M3. DBSCAN point classification

Method

1. For each point, count how many points lie within ε (include the point itself).
2. Count $\geq \text{MinPts}$ $\rightarrow$ **core**.
3. Otherwise, within ε of a core point $\rightarrow$ **border**.
4. Otherwise $\rightarrow$ **noise**.
5. Join core points within ε of each other into clusters; attach the border points.

Problem 3.1 3+5

QUESTION AS PRINTED 2082 Bhadra · Q5 · [3+5]

What are core, border and noise points? Consider the following points $(2, 10)$, $(2, 5)$, $(8, 4)$, $(5, 8)$, $(7, 5)$, $(6, 4)$, $(1, 2)$, $(4, 9)$. Use DBSCAN algorithm to find cluster with $\varepsilon = 2$ and $\text{minPts} = 2$.

The paper prints the radius as a plain e ; it is ε . Label the points $A \dots H$ in the order given, so the working can name them.

Step 1: neighbourhood of every point

$\varepsilon = 2$, and **the point counts itself**, so $\text{MinPts} = 2$ means “at least one other point within 2”.

Point	Others within $\varepsilon = 2$	$ N $	Type	Why
$A(2, 10)$	— (nearest is H at 2.236)	1	noise	no neighbour
$B(2, 5)$	— (nearest is G at 3.162)	1	noise	no neighbour
$C(8, 4)$	E (1.414), F (2.000)	3	core	≥ 2
$D(5, 8)$	H (1.414)	2	core	≥ 2
$E(7, 5)$	C (1.414), F (1.414)	3	core	≥ 2
$F(6, 4)$	C (2.000), E (1.414)	3	core	≥ 2
$G(1, 2)$	— (nearest is B at 3.162)	1	noise	no neighbour
$H(4, 9)$	D (1.414)	2	core	≥ 2

- **$C-F$ is exactly 2.000**, which is *inside* the neighbourhood: the definition is $d \leq \varepsilon$, not $d < \varepsilon$. Getting this wrong splits the cluster.
- **There are no border points here.** A , B and G are not core, and none of them is within ε of a core point either, so all three are noise. Say that explicitly — the question names all three categories.

Step 2: join the core points

- C, E, F are mutually within $\varepsilon \rightarrow$ **Cluster 1** = $\{(8, 4), (7, 5), (6, 4)\}$.
- D and H are within ε of each other and of nothing else $\rightarrow$ **Cluster 2** = $\{(5, 8), (4, 9)\}$.
- No border point is left to attach.

Final answer

- **2 clusters:** $\{(8, 4), (7, 5), (6, 4)\}$ and $\{(5, 8), (4, 9)\}$.
- **Core:** all 5 of those points. **Border:** none. **Noise:** $(2, 10), (2, 5), (1, 2)$.
- **Check:** $5 + 0 + 3 = 8$ points. ✓

Practice – the same method with MinPts = 3

PROBLEM not a PYQ · written to show the MinPts effect verified/added

Classify each of the points $A(1, 1), B(1, 2), C(2, 1), D(8, 8), E(8, 9), F(25, 25)$ as a **core**, **border** or **noise** point under DBSCAN with $\varepsilon = 2$ and MinPts = 3 (Euclidean distance, the point itself counted in its own neighbourhood). Then give the clusters DBSCAN produces.

Kept because it isolates **what MinPts alone changes**: the same shaped data gives a different answer at MinPts 3 than at MinPts 2. verified/added

Step 1: neighbourhood counts

(including the point itself)

Point	Points within $\varepsilon = 2$	Count	Type
$A(1, 1)$	A, B (1.00), C (1.00)	3	core
$B(1, 2)$	B, A (1.00), C (1.41)	3	core
$C(2, 1)$	C, A (1.00), B (1.41)	3	core
$D(8, 8)$	D, E (1.00)	2	not core
$E(8, 9)$	E, D (1.00)	2	not core
$F(25, 25)$	F only	1	noise

Step 2: form clusters

- A, B, C are all core and all within ε of each other $\rightarrow$ **Cluster 1** = $\{A, B, C\}$.
- D and E have only 2 neighbours each, and **neither is within ε of any core point** $\rightarrow$ they are **not border points** either, they are **noise**.
- F is isolated $\rightarrow$ **noise**.

Final answer

- **1 cluster:** $\{A, B, C\}$. **Noise:** D, E, F .
- **The trap:** D and E look like a cluster to the eye, but with MinPts = 3 a pair can never form one. **A border point must be within ε of an actual core point** – being near a non-core point is not enough.
- Lower MinPts to 2 and $\{D, E\}$ becomes a second cluster. **Quote your parameters with the answer.**

Ch 6 – Anomaly / Fraud Detection – Numerical Problems & Solutions

This chapter is **almost entirely theory**. Across all 23 papers there is exactly **one** numerical 76 Ch, on the base rate fallacy. It is solved below, plus 2 practice problems covering the patterns a future numerical would most likely take.

Methods

- **M1** Base rate fallacy: detection rate, false alarm rate, Bayesian detection rate
- **M2** Distance-based outlier: the $DB(r, \pi)$ test

76 Ch
practice

M1. Base rate fallacy

Method

- Let I = anomalous, $\neg I$ = normal, A = alarm raised.
- **Detection rate** = $P(A|I)$ **False alarm rate** = $P(A|\neg I)$
- **Bayesian detection rate** (what an analyst actually cares about):

$$P(I|A) = \frac{P(A|I) P(I)}{P(A|I)P(I) + P(A|\neg I)P(\neg I)}$$

- **Fastest safe route**: invent a round population (10,000), build the confusion matrix, read every rate off it. No formula to misremember.

Problem 1.1 3

QUESTION AS PRINTED 2076 Chaitra · Q11 (b) · [3]

If the probability that a normal object is classified as an anomaly is 0.01 and the probability that an anomalous object is classified as anomalous is 0.99, then what is the false alarm rate and detection rate if 99% of the objects are normal?

Given

- $P(A|\neg I) = 0.01$ (normal object raises an alarm)
- $P(A|I) = 0.99$ (anomalous object raises an alarm)
- $P(\neg I) = 0.99$, so $P(I) = 0.01$

Step 1: the two rates asked for, read straight off the givens

- **False alarm rate** = $P(A|\neg I) = 0.01$ (1%)
- **Detection rate** = $P(A|I) = 0.99$ (99%)

Step 2: why the 99% prior was given – build the matrix on 10,000 objects

	Alarm	No alarm	Total
Anomalous (I)	$0.99 \times 100 = 99$ (TP)	1 (FN)	100
Normal ($\neg I$)	$0.01 \times 9900 = 99$ (FP)	9801 (TN)	9,900
Total	198	9,802	10,000

- Check: detection rate = $99/100 = 0.99$ ✓ false alarm rate = $99/9900 = 0.01$ ✓

Step 3: Bayesian detection rate

$$P(I|A) = \frac{(0.99)(0.01)}{(0.99)(0.01) + (0.01)(0.99)} = \frac{0.0099}{0.0099 + 0.0099} = \frac{99}{198} = 0.50$$

Final answer

- False alarm rate = **0.01**, detection rate = **0.99**.
- **But the Bayesian detection rate is only 0.50** – half of every alarm raised is false, despite a detector that is 99% accurate in both directions.
- **This is the base rate fallacy**. The normal population (9,900) is so much larger than the anomalous one (100) that its 1% error rate manufactures exactly as many false alarms (99) as there are true detections (99).
- **Say this in the exam**: it is the whole point of the question and is worth more than the two arithmetic lines.

If the examiner wanted only the two conditional rates, they are given directly and the question is trivial — which is the clue that the **99% prior** must be used for something. Answer both readings: state the two rates, then compute $P(I|A)$ and name the fallacy. verified/added

Practice A — rarer anomalies

PROBLEM not a PYQ · same detector, rarer anomalies verified/added

The detector of Problem 1.1 is unchanged: a normal object is classified as an anomaly with probability 0.01, and an anomalous object is classified as anomalous with probability 0.99. Now only **1 object in 1,000** is anomalous instead of 1 in 100. Find the detection rate, the false alarm rate, and the probability that an object raising an alarm really is an anomaly.

Shows how fast the Bayesian rate collapses. verified/added

- On 100,000 objects: 100 anomalous, 99,900 normal.
- $TP = 0.99 \times 100 = 99$ $FP = 0.01 \times 99,900 = 999$
- $P(I|A) = \frac{99}{99 + 999} = \frac{99}{1098} = \mathbf{0.090}$
- **Only 9% of alarms are real.** Detection rate and false alarm rate are unchanged at 0.99 and 0.01 — **only the base rate moved.**
- **Conclusion (Axelsson):** to keep $P(I|A)$ usable when $P(I)$ is tiny you must drive the *false alarm rate* down, not push the detection rate up.

M2. Distance-based outlier: the $DB(r, \pi)$ test

Method

- Object o is a $DB(r, \pi)$ -outlier if

$$\frac{|\{o' \mid \text{dist}(o, o') \leq r\}|}{|D|} \leq \pi$$

- In words: count how many objects (including o itself, per Han & Kamber) lie within distance r ; if that count as a fraction of $|D|$ is at most π , it is an outlier.
- Equivalent form: with $k = \lceil \pi |D| \rceil$, o is an outlier if $\text{dist}(o, o_k) > r$.

Practice B

PROBLEM not a PYQ · written to fill the gap verified/added

A dataset contains the points $P_1(1, 1)$, $P_2(1, 2)$, $P_3(2, 1)$, $P_4(2, 2)$ and $P_5(8, 8)$. Taking $r = 2$ and $\pi = 0.4$, and using Euclidean distance, determine which points are $DB(2, 0.4)$ -outliers.

No PYQ has set this yet, but it is the only computable form of the distance-based question that is asked in 6 papers, so it is worth one run-through.

verified/added

Step 1: distance matrix

	P_1	P_2	P_3	P_4	P_5
P_1	0	1	1	1.414	9.899
P_2	1	0	1.414	1	9.220
P_3	1	1.414	0	1	9.220
P_4	1.414	1	1	0	8.485
P_5	9.899	9.220	9.220	8.485	0

Step 2: count neighbours within $r = 2$

(threshold $\pi |D| = 0.4 \times 5 = 2$ objects)

- P_1 : itself, $P_2, P_3, P_4 = 4$ $4/5 = 0.8 > 0.4 \rightarrow$ normal
- P_2, P_3, P_4 : 4 each by symmetry $\rightarrow$ normal
- P_5 : only itself = 1 $1/5 = 0.2 \leq 0.4 \rightarrow$ **outlier**

Final answer

P_5 is the only $DB(2, 0.4)$ -outlier.

- **Cross-check with the k -NN form:** $k = \lceil 0.4 \times 5 \rceil = 2$. For P_5 , its 2nd nearest neighbour is at $8.485 > 2 \rightarrow$ outlier $\checkmark$. For P_1 , its 2nd nearest is at $1 \leq 2 \rightarrow$ normal $\checkmark$.

- **Note the sensitivity to r :** raise r to 9 and P_5 stops being an outlier altogether. Quoting the chosen r and π is part of the answer.

Ch 7 – Advanced Applications – Numerical Problems & Solutions

This chapter is almost entirely short notes. Across all 23 papers there is exactly **one** numerical 76 Ch, on PageRank. It is solved in full below, plus 1 practice problem on the variant most likely to be set next.

M1. PageRank

Method

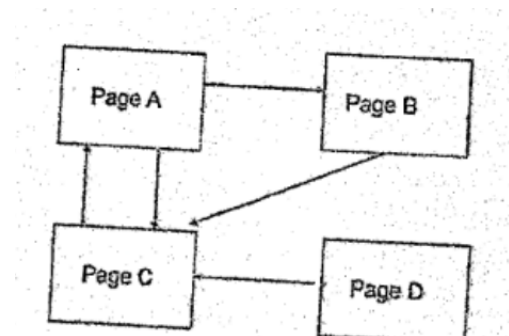
$$PR(A) = (1 - d) + d \sum_i \frac{PR(T_i)}{C(T_i)}$$

1. Write down the link table first: for every page, its **outlinks** (to get C) and its **inlinks** (to know which terms to sum). **Nearly every error comes from confusing the two.**
2. Initialise all ranks (here 0.25).
3. Each iteration, compute all pages using the **previous** iteration's values, then substitute in together.
4. Repeat for the required number of iterations, or until the values settle.
 - A page's own rank is split equally among its outlinks – that is the $/C(T_i)$.
 - A page with **no inlinks** sits permanently at $(1 - d) = 0.15$.

Problem 1.1 8

QUESTION AS PRINTED 2076 Chaitra · Q12 · [8]

Consider the following subset of pages and their links. Apply the PageRank algorithm using a damping factor of 0.85. A minimum of five iterations are required. Assume initial page rank of all pages is 0.25.



Step 1: read the graph

Page	Outlinks	C	Inlinks
A	B, C	2	C
B	C	1	A
C	A	1	A, B, D
D	C	1	none

Step 2: write the 4 equations

($d = 0.85$, $1 - d = 0.15$)

- $PR(A) = 0.15 + 0.85 \frac{PR(C)}{2}$
- $PR(B) = 0.15 + 0.85 \frac{PR(A)}{1}$
- $PR(C) = 0.15 + 0.85 \left(\frac{PR(A)}{2} + PR(B) + PR(D) \right)$
- $PR(D) = 0.15 + 0.85(0) = \mathbf{0.15}$ **constant, D has no inlinks**
- Note A's rank is divided by 2 because A links to both B and C.

Final answer after 5 iterations

- $PR(A) = 0.972$, $PR(B) = 0.519$, $PR(C) = 1.028$, $PR(D) = 0.150$

Step 3: iterate

Iter	PR(A)	PR(B)	PR(C)	PR(D)
0	0.2500	0.2500	0.2500	0.2500
1	0.3625	0.2563	0.6813	0.1500
2	0.7291	0.3041	0.6494	0.1500
3	0.7020	0.4599	0.8459	0.1500
4	0.8690	0.4484	0.9668	0.1500
5	0.9718	0.5193	1.0280	0.1500

Row 0 is the given start value.

Sample working (iteration 1)

- $PR(A) = 0.15 + 0.85(0.25) = 0.3625$
- $PR(B) = 0.15 + 0.85(0.25/2) = 0.2563$
- $PR(C) = 0.15 + 0.85(0.125 + 0.25 + 0.25) = 0.6813$
- $PR(D) = 0.15$

– Ranking: **C > A > B > D**

Two comments worth writing, they show understanding. verified/added

1. The values are still climbing at iteration 5 but the **ranking has been stable since iteration 3**. PageRank converges on *order* much faster than on *value*, which is why 5 iterations is enough for the question.
2. Solving the equations exactly gives the fixed point $PR(A) = 1.490$, $PR(B) = 0.783$, $PR(C) = 1.577$, $PR(D) = 0.150$, and these sum to **4.00 = N**. **That sum is your check** – in this (non-normalised) form the ranks always converge to the number of pages.
3. **C wins** because it collects links from three pages while A collects only one, and **D is stuck at 0.15** because nothing points to it. Linking out never raises your own rank.

Practice – the normalised form

PROBLEM not a PYQ · the same graph, normalised form verified/added

Repeat Problem 1.1 on the same four-page graph, but using the *normalised* PageRank formula

$$PR(A) = \frac{1-d}{N} + d \sum_i \frac{PR(T_i)}{C(T_i)}$$

with $d = 0.85$, $N = 4$ and an initial rank of 0.25 for every page. Compare the converged ranks and the ordering with those of Problem 1.1.

This form makes the ranks sum to 1 instead of N . Know both. verified/added

- With $N = 4$, $\frac{1-d}{N} = \frac{0.15}{4} = 0.0375$.
- Iteration 1 from the same 0.25 start:
 - $PR(A) = 0.0375 + 0.85(0.25) = 0.2500$
 - $PR(B) = 0.0375 + 0.85(0.125) = 0.1438$
 - $PR(C) = 0.0375 + 0.85(0.625) = 0.5688$
 - $PR(D) = 0.0375$
- Sum = 1.0001 ≈ 1 ✓ – **that is the check for this form.**
- **The ranking is identical** (C > A > B > D); only the scale differs.
- **In the exam:** state which form you are using in one line. Either is accepted; mixing them is not.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity–relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.